Who Revises First?

Analyst Revision Flow and Zero-Beta Stock Selection in Concentrated Equity Portfolios

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Abstract. Concentrated equity portfolios, such as sector and thematic funds, high-conviction books and systematic baskets on narrow universes, face a statistical problem that diversified quantitative portfolios do not. With few names and a dominant common factor, stock-selection alpha is hard to measure, easy to confuse with beta and momentum, and readily manufactured by risk overlays. Using point-in-time sell-side target prices, I show that a first-generation regime-timed strategy reaches a development Calmar ratio of 1.77 through market exposure rather than selection. I then propose *First-Mover Flow* (FMF), a zero-beta book whose key ingredient, *leader minus herd*, separates revisions that break an information silence from copycat revisions that follow earnings clusters. Traded through horizon-matched overlapping tranches, FMF earns a net Sharpe ratio of 1.13 and a factor-adjusted alpha of 7.4% per year ($t = 2.98$) over 2017–2021. On 30 stocks over a never-seen period (2024–2026), the frozen, pre-registered specification earns a Sharpe ratio of 1.05 and a Calmar ratio of 1.32 at zero beta, and leader minus herd replicates. I then attack the result with nine objections. The edge comes from timing, not from a hindsight tilt (causal timing book $t = 2.41$, and $t = 2.21$ out of sample). It survives execution delay, borrow fees and costs (breakeven 234 bp). 99% of some 300 alternative specifications earn a positive Sharpe ratio. A Bayesian random-effects meta-analysis across three independent samples gives a 0.96 probability that the signal works in the next universe. A hybrid GARCH-diffusion generator, which passes the discriminative and memorisation tests that TimeGAN fails, turns the single in-sample Calmar ratio into a posterior-predictive distribution and identifies momentum crashes as the binding stress.

Keywords: concentrated portfolios, stock selection, analyst target prices, herding, first movers, Calmar ratio, backtest overfitting, multiverse analysis, Bayesian meta-analysis, diffusion models, stress testing.
JEL codes: G11, G12, G14, G17, G23, G24.

[1]Code, data contract, tests and every number in this paper are reproducible from the accompanying open-source repository (`https://github.com/bahakh19/first-mover-flow`, Python package `fmflow`) and its research notebook. The out-of-sample universe of Section 9 and its pre-registration file are distributed with the repository. Errors are my own.

1 Introduction

A large share of active equity capital is managed in *concentrated* portfolios: technology and thematic funds, sector specialists, high-conviction "best ideas" books, and systematic baskets built on narrow universes for structured products. These managers are judged on drawdown-sensitive metrics. Their capital is gated by allocators on peak-to-trough losses, performance fees are subject to high-water marks, and structured overlays are priced off realised volatility and drawdown. The natural objective is therefore the ratio of compounded return to maximum drawdown, the Calmar ratio, rather than the Sharpe ratio alone.

Concentration creates three problems that the literature on diversified factor portfolios largely sidesteps. The first is *breadth*. By the fundamental law of active management (Grinold, 1989; Clarke, de Silva, and Thorley, 2002), the information ratio scales with the square root of the number of independent bets, and a ten-stock universe offers very few. The second is *factor dominance*. In a concentrated universe a single market or sector factor explains most of the variance (the average pairwise correlation in the development universe below is 0.40), so almost any signal correlates with beta or momentum. The third is *camouflage*. Overlays that scale net exposure with a regime model or a trend filter can lift the Calmar ratio of a weak selection book through market beta, especially in a rising market, and the resulting statistics look like skill. These three problems interact: low breadth means that the selection component is measured with little precision, which makes it easy for the overlay to dominate the reported performance.

This paper asks a practical and an academic question. Practically: *can a concentrated manager extract a stock-selection alpha that is genuinely independent of market exposure, and improve the Calmar ratio through that alpha rather than through beta?* Academically: *how should such an alpha be validated when the sample is too small for conventional significance tests to have power and when many specifications have been tried?*

I answer both questions with sell-side target prices, a data source that is available point in time for every listed large cap and whose information content is well documented (Brav and Lehavy, 2003; Asquith, Mikhail, and Au, 2005). The research proceeds in three stages, each reported in full.

Stage 1: a strategy that solved the engineering and missed the alpha: A first-generation strategy, *Regime-Timed Breadth* (RTB), combines a revision-breadth composite with GJR-GARCH risk weighting and a hidden-Markov regime tilt. It is built with strict point-in-time discipline and reaches a development Calmar ratio of 1.77. Its decomposition is sobering. The timing leg earns 11.0% a year, the selection leg reaches a Calmar ratio of only 0.36, timing regressions in the spirit of Treynor and Mazuy (1966) and Henriksson and Merton (1981) find no timing ability, and a static exposure of the same average size beats the regime tilt. The headline is beta (Figure 1).

Stage 2: a new signal and a horizon-matched construction: I propose *leader minus herd* (LMH), which classifies each revision by its arrival time. Leaders revise on an information-event day or after at least ten sessions without any other revision on the stock, consistent with private research effort (Cooper, Day, and Lewis, 2001). Herders revise 3–15 sessions after an

earnings cluster, in its direction, consistent with career-concern herding (Welch, 2000; Hong, Kubik, and Solomon, 2000; Clement and Tse, 2005). Leader breadth predicts returns (rank IC 0.066), herder breadth predicts them negatively (-0.084), and LMH is the only one of 18 candidate signals with incremental power beyond four price factors (Fama–MacBeth $t = 2.46$). Combined with three breadth signals in an equal-weight rank composite and traded through 42 daily overlapping tranches, the zero-beta strategy *First-Mover Flow, market-neutral* (FMF-N) earns a net Sharpe ratio of 0.98 in development, 1.33 in the 2020–21 holdout and 1.13 overall. It roughly doubles the selection Calmar ratio of RTB at zero beta.

Stage 3: adversarial validation:

- **Institutional comparables:** FMF-N is compared with 13 signals and 3 allocations that institutional desks run. It has the highest Sharpe and Calmar ratios among the cross-sectional books built identically, returns $+12.0\%$ in the COVID crash while the market returns -35.5%, carries an alpha of 6.1% ($t = 2.51$) against the comparables not built from its own components, and improves 16 of 18 books when blended.
- **Concentration:** A full leave-one-stock-out rebuild shows that no single name carries the result.
- **A never-seen period:** Most importantly, the frozen specification is applied, under a hashed pre-registration, to 30 stocks over 2023–2026, a period never used for any choice; twenty of the names are new and ten are the development names. It earns a Sharpe ratio of 1.05, a Calmar ratio of 1.32 and an alpha t-statistic of 2.04 at zero beta, ranks 2nd of 14 cross-sectional books on both metrics, and LMH replicates with essentially the same information coefficient.

Stage 4: attacking the result: Section 10 states nine objections that were raised during the analysis of the strategy and answers each with a test chosen to break the result.

- **Hindsight:** A causal split of the book into a static tilt and a timing component shows that the edge is timing (Sharpe 1.04 in development, 1.33 out of sample). The static tilt towards stocks that later did well is insignificant, and it loses money out of sample.
- **Implementation:** A one-session delay, a 100 bp borrow fee and 10 bp costs leave a Sharpe ratio of 1.06.
- **Research degrees of freedom:** A research space of about 300 specifications, including the quarterly target expiry an investment committee would ask about, keeps the sign in 99% of cases. The selection-free ensemble earns 0.92.
- **The new signal:** Adding leader minus herd to classic breadth raises the Sharpe ratio by 0.28 ($p = 0.028$) in sample, and by a similar amount out of sample.
- **Small samples:** Bayesian posteriors under sceptical priors, a random-effects meta-analysis across three independent samples, and empirical-Bayes shrinkage of the signal search replace single p-values with the probabilities an allocator needs.

Section 11 turns the Calmar ratio from a single draw into a distribution. Its generative models include a hybrid of GJR-GARCH dynamics and a conditional diffusion model for the joint

shocks.

The paper makes three contributions. The first is economic: the timing of analyst revisions, not only their direction, carries information, and in concentrated universes it delivers a zero-beta alpha that blends well with the factor strategies institutions already run. The second is methodological: a validation protocol for small, heavily searched samples. It combines calibrated non-overlapping tests, persistence-preserving placebos, deflation over the full research log (Bailey and López de Prado, 2014; Bailey et al., 2017), a multiverse with family-wise tests (Hansen, 2005; Romano and Wolf, 2005), Bayesian meta-analysis across samples, pre-registration and replication. The third is practical: an open, tested Python package. It reproduces every number, and it runs the same protocol, a production book, a QIS index and generative stress tests on any user's universe. The protocol also keeps the conclusions honest. With ten stocks only information coefficients above about 0.138 are detectable. The development search, taken alone, is suggestive rather than conclusive. Momentum crashes are a genuine risk of the book. I report these limits alongside the results, together with a ledger of more than fifty signals and hypotheses that failed.

The remainder is organised as follows. Section 2 reviews the literature. Section 3 formalises the three problems and the validation protocol. Section 4 describes the data. Section 5 analyses the first-generation strategy. Section 6 defines First-Mover Flow. Section 7 reports development and holdout results, Section 8 the comparison with institutional strategies, and Section 9 the out-of-sample replication. Section 10 confronts the result with nine objections, and Section 11 studies path risk with generative models. Section 12 draws the implications for practitioners, and Section 13 concludes.

2 Related literature

Analyst revisions and target prices: Earnings-forecast revisions predict returns (Stickel, 1991), revision-based and price momentum are related but distinct (Chan, Jegadeesh, and Lakonishok, 1996), the market under-reacts to less visible revisions (Gleason and Lee, 2003), and recommendation *changes* carry information that overlaps with momentum (Jegadeesh et al., 2004). Target prices move prices (Brav and Lehavy, 2003; Asquith, Mikhail, and Au, 2005), but analysts' ability to forecast target-price levels is limited and does not persist (Bradshaw, Brown, and Huang, 2013), and relative valuation within an industry is more informative than absolute upside (Da and Schaumburg, 2011). Consistent with the latter two results, signals based on target-price *levels* fail in this paper, while signals based on the *flow* of revisions succeed.

Leaders, herders and disagreement: Cooper, Day, and Lewis (2001) identify lead analysts by the timing of their forecasts and show that their revisions have more price impact. Herding models and evidence (Welch, 2000; Hong, Kubik, and Solomon, 2000; Clement and Tse, 2005) imply that revisions issued close to others carry less information. LMH turns this timing distinction into a stock-level cross-sectional signal. Diether, Malloy, and Scherbina (2002) relate dispersion of opinion to returns, which motivates one of the comparables.

Institutional factor strategies: The comparables follow the standard toolkit: cross-sectional momentum (Jegadeesh and Titman, 1993), short-term reversal, low volatility and betting-against-beta (Frazzini and Pedersen, 2014), residual momentum (Blitz, Huij, and Martens, 2011), 52-week-high proximity (George and Hwang, 2004) and time-series momentum (Moskowitz, Ooi, and Pedersen, 2012). Volatility management (Barroso and Santa-Clara, 2015; Moreira and Muir, 2017) and regime filters (Hamilton, 1989) are the standard overlays.

Construction, costs and inference: Overlapping holding-period portfolios go back to Jegadeesh and Titman (1993). Trading costs eliminate many anomalies (Novy-Marx and Velikov, 2016), and estimation error favours naive weights (DeMiguel, Garlappi, and Uppal, 2009). On inference, Harvey, Liu, and Zhu (2016) argue for higher hurdles. I use the probabilistic and deflated Sharpe ratios (Bailey and López de Prado, 2012, 2014), the probability of backtest overfitting (Bailey et al., 2017), Holm's correction (Holm, 1979), the stationary bootstrap (Politis and Romano, 1994), Sharpe-difference tests (Jobson and Korkie, 1981; Memmel, 2003), Newey–West and Fama–MacBeth inference (Newey and West, 1987; Fama and MacBeth, 1973), asymmetric GARCH (Glosten, Jagannathan, and Runkle, 1993), and purged evaluation (López de Prado, 2018).

Data snooping, small samples and generative scenarios: White (2000), Hansen (2005) and Romano and Wolf (2005) provide bootstrap tests that account for the search over many strategies; Ledoit and Wolf (2008) give a robust test of Sharpe-ratio differences. The multiverse (Steegen et al., 2016) and specification-curve (Simonsohn, Simmons, and Nelson, 2020) approaches report results over all reasonable analytic choices. Lo (2002) corrects Sharpe ratios for serial correlation and Lo (2008) separates the active and passive components of a strategy's return. On the Bayesian side, I use the Bayesian bootstrap (Rubin, 1981), hierarchical models with weakly informative priors on variance components (Gelman, 2006) and empirical-Bayes shrinkage (Efron, 2010); McLean and Pontiff (2016) motivate sceptical priors through the post-publication decay of anomalies. For path simulation, filtered historical simulation (Barone-Adesi, Giannopoulos, and Vosper, 1999) with variance targeting (Engle and Mezrich, 1996), which is the industry standard. Deep generative models for financial time series include GANs (Wiese et al., 2020) and TimeGAN (Yoon, Jarrett, and van der Schaar, 2019). Denoising diffusion models (Ho, Jain, and Abbeel, 2020; Nichol and Dhariwal, 2021; Song, Meng, and Ermon, 2021) have become the leading generative technique in other domains. I combine the econometric and the diffusion approaches in a way that suits small samples.

3 Why concentrated universes are hard: a framework

3.1 Breadth and detectability

Let IC_t be the daily cross-sectional Spearman correlation between a signal and forward h-session returns across N stocks. Under the null, its standard deviation is approximately $1/\sqrt{N-1}$. With M independent (non-overlapping) observations, the smallest true IC de-

tectable with power $1-\beta$ in a one-sided test at level α is

$$\mathrm{MDE} = \frac{t_{1-\alpha,M-1} + t_{1-\beta,M-1}}{\sqrt{N-1}\,\sqrt{M}}. \tag{1}$$

For $N = 10$ and $h = 21$, a three-year sample ($M \approx 35$) gives an MDE of about 0.14, and six years about 0.10. For $N = 100$ the same samples give 0.04 and 0.03. Published single-anomaly ICs are typically 0.01–0.05. In a concentrated universe a genuinely good signal therefore sits near the detection boundary by construction, and we must combine evidence from several directions instead of relying on one test.

3.2 Beta camouflage

Consider a book with long and short legs of budgets $\frac{1}{2} \pm \tau_t$, where τ_t is a regime-dependent tilt. Its return decomposes into a dollar-neutral selection component and a timing component with net exposure $2\tau_t$:

$$r^{\mathrm{book}}_{t+1} = \underbrace{\tfrac{1}{2}\left(r^{L}_{t+1} - r^{S}_{t+1}\right)}_{\text{selection}} + \underbrace{\tau_t\left(r^{L}_{t+1} + r^{S}_{t+1}\right)}_{\text{timing} \approx 2\tau_t\, r^{m}_{t+1}}. \tag{2}$$

An overlay adds value only if the timing component beats a *static* exposure of the same average size, $2\bar{\tau}\, r^m$, and if timing regressions detect ability. Otherwise its contribution to Sharpe and Calmar is the equity premium of the sample. I apply this test to every overlay in the paper.

3.3 The Calmar ratio as an objective

Calmar is the natural objective for drawdown-constrained capital, but it is a noisy statistic: in a three-year sample the maximum drawdown is set by one or two episodes, and the stationary-bootstrap 5–95% interval of a selection book's Calmar spans roughly -0.1 to 2.5. Choosing among specifications by point Calmar mostly chooses among drawdown paths. I therefore select by the *median* Calmar across stationary-bootstrap resamples of the development sample (Politis and Romano, 1994). I report the probability that this choice is overfit, and I treat market exposure as a separate, explicit decision (a beta budget) that is never selected by in-sample Calmar.

3.4 A validation protocol for small, searched samples

1. **Point-in-time construction with a truncation test:** Every feature computed on data truncated at T must equal the full-sample feature up to T. Negative controls, such as centred windows or full-sample demeaning, must fail the test.
2. **Calibrated inference:** With h-session labels sampled daily, the Newey–West t with h lags over-rejects (about 9% at a nominal 5% on this date grid). I use a phase-averaged non-overlapping test, whose simulated size is 4–5%, and Holm's correction across families.
3. **Persistence-preserving placebos:** IID random scores are a weak null for a persistent signal. I add a stock-label permutation, which assigns each stock's real signal history to

another stock, and a circular time-shift of the signal panel. The time-shift also preserves each stock's habitual rank, so it asks whether *timing* adds value beyond stock identity.

4. **Deflation over the research log:** Deflated Sharpe ratios count every configuration ever examined, using the log's actual correlation structure, and the probability of backtest overfitting is computed for each selection step.
5. **Pre-registration and replication:** Hypotheses and transfer rules are hashed before outcomes are computed, and the final test uses a period that was never used for any choice.

4 Data

Both universes share the same format: daily closing prices normalised to one at the first observation, and analyst target-price records (publication date, stock, analyst, target on the same scale as the price).

Development universe: Ten US large caps, January 2016 to December 2021 (1,511 sessions after cleaning). There are 5,929 accepted target prices from 304 analysts: 3,168 up revisions, 1,020 down revisions, 1,260 reiterations and 481 initiations. The first year is a warm-up. Development (DEV) is 2017–2019; the holdout is 2020–2021. Table 1 describes the cross-section. It is concentrated and heterogeneous: total returns range from 1.4 to 36.3 times, volatilities from 27% to 53%, and coverage from 17 to 72 analysts per stock.

Open universe, never-seen period: 30 US stocks from January 2023 to August 2026 (899 sessions), with 2,363 accepted target prices from 50 brokers and 109 event clusters. The development data are a proprietary, analyst-level dataset; the open data are public, with broker-level target prices from a Kaggle dataset[2] and closing prices from Yahoo Finance, normalised in the same way (see `DATA_SOURCES.md` and the `Data_Gathering` notebook in the repository). The period starts two years after the development data end. The universe consists of the ten development names and twenty large caps that were never used in development; Section 9 checks that the overlap does not drive the result. It differs structurally from the development universe: there are three times as many names; coverage is far sparser, with 13 analysts per stock at the median, most covering several of the stocks (median 6); total returns range from 0.04 to 14.9 times; and the average pairwise correlation is 0.26.

Point-in-time construction: Prices are cleaned causally: non-sessions (dates on which fewer than half the stocks print) are removed, and isolated gaps are forward-filled for at most five sessions and never backward-filled. A target published on day t_{pub} becomes known at the next session and expires after 252 sessions. Implausible targets (target/price outside $[0.2, 5]$) and revisions inconsistent with the price move since the analyst's previous target are removed (Appendix A). Each record is classified as an initiation, a reiteration, or a fresh up or down

[2] `https://www.kaggle.com/datasets/yijian2427/us-market-analyst-insights-with-full-stock-symbols` (MIT licence).

Table 1. Development universe

Stock	Total return (×)	Ann. vol (%)	Analysts	Ratings	Fresh revisions	Event clusters	Median active (DEV)
stock_1	6.74	29.4	72	775	588	25	30
stock_2	1.43	44.1	31	430	325	5	15
stock_3	4.92	52.6	17	240	162	17	8
stock_4	1.52	33.0	50	639	401	29	25
stock_5	2.49	29.2	29	463	355	2	18
stock_6	6.14	26.8	58	670	467	24	25
stock_7	2.71	28.2	44	620	418	27	24
stock_8	36.35	47.1	54	801	616	30	27
stock_9	3.25	36.9	54	810	528	30	30
stock_10	3.65	36.4	36	481	328	25	17

Fresh revisions are target changes by an analyst with an active previous target on the stock. Event clusters are (stock, publication date) pairs with at least five ratings. Median active is the median number of analysts with a valid target (at most 252 sessions old) during 2017–2019.

revision. Every feature at session t uses closes up to t and targets with knowledge date $\leq t$, which is verified by the truncation test.

5 The first-generation strategy and what it left unsolved

Design: RTB ranks stocks on an equal-weight composite of three revision-breadth features (63-session breadth, event-day breadth and momentum-orthogonal breadth). It is long the top three and short the bottom three, rebalances weekly with hysteresis, allocates within legs by inverse GJR-GARCH-t volatility, and tilts the leg budgets by ± 0.25 when a forward-filtered two-state hidden Markov model signals a calm regime and the equal-weight index is above its 200-day average.

What it solved: RTB established the infrastructure that every later result relies on: point-in-time target snapshots with knowledge-date stamping, a documented cleaning audit, truncation tests, walk-forward risk models, a frozen specification and a vectorised backtest. Its development performance was strong: CAGR 16.6%, Sharpe 1.56, Calmar 1.77.

What it did not solve: Figure 1 decomposes RTB.

- **Timing:** The timing leg earns 11.0% a year at a Sharpe ratio of 1.69. Its average net exposure is $+0.35$, and a static exposure of that size (0.35 times the market) ends above the timing leg. Treynor–Mazuy and Henriksson–Merton regressions find no timing ability.
- **Selection:** The selection leg earns a Sharpe ratio of 0.63 and a Calmar ratio of 0.36 in DEV.
- **Signal:** Its breadth features are 0.55–0.78 rank-correlated with price momentum.

In the language of Section 3, RTB's Calmar ratio was camouflaged beta on top of a momentum-like selection signal.

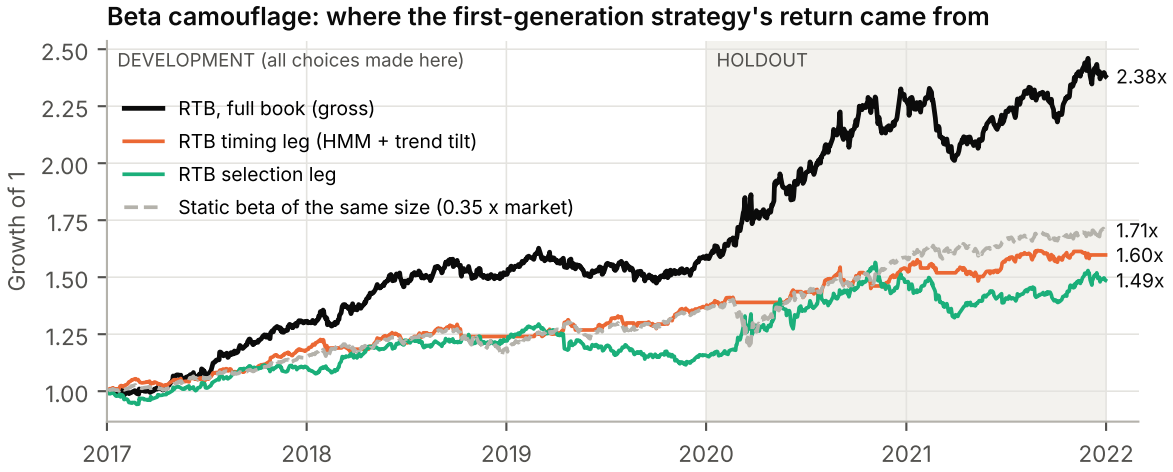


Figure 1. Beta camouflage in the first-generation strategy (RTB). Cumulative gross returns of the full book, its timing leg (the hidden-Markov and trend tilt of the leg budgets), its dollar-neutral selection leg, and a static market exposure equal to the book's average net exposure (0.35). The static exposure ends above the timing leg: the overlay's contribution is the equity premium of the sample, not timing skill.

The search that followed: Six audit rounds then tried to find a selection alpha, pre-registering every new family of hypotheses (Table 2). The rounds tested the following, and none passed its pre-registered tests:

- novel analyst features, an extended library of 18 signals and signal blends
- analyst-network spillovers, idiosyncratic momentum and a Lévy-area lead–lag signal
- event-level post-revision drift, analyst-skill weighting and earnings-cycle conditioning
- within-analyst relative valuation, normalised revision velocity and dispersion gating
- earnings-reaction predictability, plus risk-forecasting overlays (Round 6).

Two lessons generalise. Conditioning on subsets of the sample (event days, analyst histories, cycle phases) destroys more power through lost breadth than it gains in precision. Signals built on target-price *levels* are contaminated by the price in their denominator. The remaining problem was precise: a selection signal that adds to price momentum, and a construction that harvests it without leaning on beta.

6 First-Mover Flow: signal and construction

Let $U_{i,s}$ and $D_{i,s}$ count the fresh up and down revisions on stock i that become known at session s.

Revision-flow signals: *Breadth* over the last 63 sessions (one earnings cycle) is

$$B_{i,t} = \frac{\sum_{s=t-62}^{t}(U_{i,s} - D_{i,s})}{\sum_{s=t-62}^{t}(U_{i,s} + D_{i,s})}, \tag{3}$$

Table 2. Research ledger: hypotheses tested before and around the final specification

Round	Hypotheses / signals	Outcome
1	Fresh-vs-stale consensus wedge, leader/laggard revisions, conviction shock (target change net of price change), post-event drift, bold-revision breadth, peer spillover; quarterly target expiry	None passed the inclusion rule; strongest (conviction shock) IC21 0.034, $p = 0.30$; a 252-session expiry is retained.
2	Library of 18 signals (11 analyst, 7 price); four pre-specified blends; analyst window; rebalancing frequency	LMH is the only signal with FM $t > 2$; blends do not improve; flow window 63 sessions; slower rebalancing dominates.
3	Analyst-network pending revisions (2), idiosyncratic momentum, Lévy-area lead–lag, upside per unit of volatility; tranche period; volatility scaling	0 of 10 significant after Holm; network signals wrong-signed; slow tranches; volatility scaling hurts.
4	Eight pre-registered tests: event-level LMH, post-revision drift, confirmed surprise, analyst-skill persistence, two skill-weighted breadths, earnings-phase interaction and gating; overlay decomposition	0 of 8 after Holm; skill does not persist ($t = -0.22$); event drift reverses in DEV but not in a one-shot holdout test; the overlay is beta.
5	Within-analyst demeaned and ranked target returns, top-pick-weighted breadth, normalised revision velocity (2), dispersion-gated breadth, quiet-day breadth, coverage change, breadth acceleration; linear-rank weights	0 of 9; within-analyst valuation wrong-signed (IC21 -0.089); flow variants duplicate breadth (correlation 0.86–0.96); linear-rank weights worse.
6	Feasibility, persistence-preserving placebos, signal decomposition, earnings-reaction predictability, variance forecasting, risk-based sizing	Reaction predictability null in DEV; analyst dispersion and an earnings-calendar proxy forecast variance beyond GARCH, but sizing by them does not improve the book.

The audit also caught and corrected a construction error: a rank-proportional book that was not dollar-neutral (a cross-sectional rank centred at $0.5/N$ rather than zero), exposed by its beta and by its random-score placebo.

defined with at least three revisions. Using the sign rather than the size of revisions is robust to the heavy tails of revision magnitudes. *Event breadth* applies the same ratio to revisions published on information-event days (at least five same-day ratings). *Momentum-orthogonal breadth* is the residual of the cross-sectional rank of B on the rank of 6-1 month momentum, date by date.

Leader minus herd: For each fresh revision r on stock i with direction $d_r = \operatorname{sign} \Delta \ln TP$:

- r is a *leader* if it is published on an event day, or if no other fresh revision on the stock occurred in the previous ten sessions
- r is a *herder* if it is not a leader, arrives 3–15 sessions after the stock's latest event cluster, and points in the cluster's direction.

Over 126 sessions (two earnings cycles, so that the sparse events cover almost every stock-day),

$$\mathrm{LMH}_{i,t} = \frac{\sum_{r \in \mathcal{W}_t} d_r \, \mathbb{1}_r^{\mathrm{lead}} - \sum_{r \in \mathcal{W}_t} d_r \, \mathbb{1}_r^{\mathrm{herd}}}{\#\{\text{leaders}\} + \#\{\text{herders}\}}, \tag{4}$$

defined with at least three classified revisions. All classifications use only revisions known at t.

Composite: The score is the equal-weight average of the four cross-sectional percentile ranks, so no weights are estimated (DeMiguel, Garlappi, and Uppal, 2009).

Construction and risk logic:

- **Daily target:** Each session a target book is formed: long the k highest-ranked and short the k lowest-ranked names. Each leg holds half of gross capital, allocated in proportion to $1/\hat{\sigma}_{i,t}$, where $\hat{\sigma}_{i,t}$ is a walk-forward GJR-GARCH(1,1)-t volatility forecast (Glosten, Jagannathan, and Runkle, 1993).
- **Held book:** The book actually held is the average of the last f targets,

$$w_{i,t} = \frac{1}{f} \sum_{s=t-f+1}^{t} w_{i,s}^{\text{target}}, \tag{5}$$

  which is identical to holding f overlapping sub-portfolios, one opened each session (Jegadeesh and Titman, 1993). It removes rebalancing-anchor luck (with a single weekly anchor the development Sharpe of the same signal ranges from 0.34 to 0.71) and spreads execution.
- **Exposure:** The book is dollar-neutral at every session. Market exposure, if a mandate wants it, is a separate, constant and disclosed beta budget c (*FMF-B20* for $c = 0.2$), never a timed overlay.
- **Costs:** Net returns are charged 5 bp per unit of traded notional.

The construction parameters ($k = 4$, $f = 42$, inverse-volatility weights) are selected on DEV from a 30-book grid by bootstrap-median Calmar (Section 7.2).

7 Development and holdout results

7.1 Signal evidence and mechanism

Table 3 and Figure 2 report the development evidence. All four signals have positive IC at every horizon. The composite reaches an IC21 of 0.109 (hit rate 62%, calibrated $p = 0.045$). LMH has the highest IC21 among the components, is positive in all three years, and is the only signal whose Fama–MacBeth t-statistic exceeds two after the four price controls. The breadth signals lose significance once momentum is controlled for, confirming that breadth alone is largely momentum. The mechanism is visible directly: leader breadth predicts returns, herder breadth predicts reversals, and their difference combines both effects. Information builds over weeks: the composite's IC rises from 0.026 at one session to 0.109 at 21.

7.2 Construction and the price of the search

The grid ($k \in \{3, 4, 5\}$, $f \in \{21, 42, 63, 84, 126\}$, equal or inverse-volatility weights) is summarised in Table 4 and Figure 3. The pre-declared rule selects $k = 4$, $f = 42$ and inverse-volatility weights, with a bootstrap-median Calmar of 0.855, against 0.735 for the equal-weight 63-session book. The diagnostics show that this choice is largely cosmetic. The 30 books are near-duplicates (average correlation 0.91), their in-sample ranking does not persist (PBO 0.945), and what persists is the level: the in-sample winner's median out-of-sample Sharpe is 0.66. The deflated Sharpe ratio is 0.94 against the grid, 0.38 against the research log of 71

Table 3. Signal evidence, development sample (2017–2019)

	IC5	IC21	IC63	Hit (%)	t_{NW}	t_{phase}	p	Holm	Years > 0	FM t
breadth_63	0.036	0.072	0.064	61	1.50	1.16	0.127	0.30	2/3	0.21
event_breadth_63	0.033	0.085	0.082	58	1.73	1.26	0.108	0.30	2/3	0.15
breadth_resid_mom	0.045	0.083	0.109	56	2.03	1.45	0.078	0.30	2/3	0.36
leader_minus_herd	0.049	0.095	0.104	60	1.81	1.47	0.075	0.30	3/3	2.46
Composite	0.053	0.109	0.114	62	2.19	1.74	0.045	–	2/3	0.60
leaders only	0.039	0.066	0.051	57	1.28	1.03	0.155	–	2/3	0.85
herders only	−0.123	−0.084	−0.148	39	−0.78	–	–	–	0/3	−0.10

Mean daily Spearman rank IC with purged forward log returns. t_{NW}: Newey–West with 21 lags (shown for continuity, over-rejects). t_{phase}, p: calibrated phase-averaged non-overlapping test, one-sided. Holm: adjusted over the four components. FM t: Fama–MacBeth t controlling for 12-1 and 6-1 momentum, 1-month reversal and 60-day volatility.

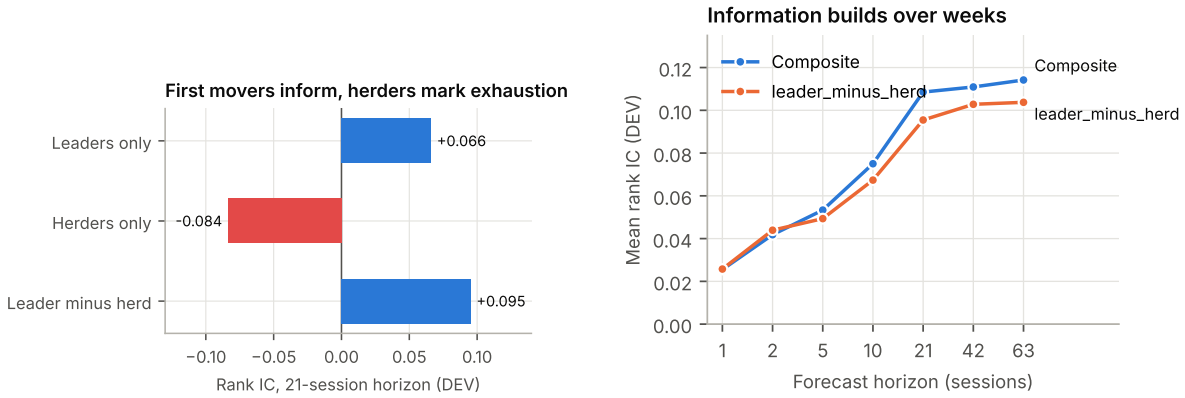


Figure 2. Mechanism and information horizon. Left: rank IC (21 sessions) of breadth computed from leader revisions only, herder revisions only, and their difference. Right: mean development IC of the composite and of LMH by horizon.

Table 4. Development construction grid (top eight books)

k	f	Weights	Sharpe	CAGR (%)	MaxDD (%)	Calmar	Boot. median	Boot. 5%	Turnover
4	42	IV-GJR ♦	0.98	7.7	−9.7	0.79	0.85	−0.02	2.1
5	42	IV-GJR	0.96	6.3	−7.7	0.81	0.83	0.00	1.9
5	63	IV-GJR	0.96	6.1	−7.5	0.81	0.81	0.00	1.4
4	63	IV-GJR	0.95	7.3	−9.6	0.76	0.77	−0.05	1.7
4	42	EW	0.91	8.1	−12.3	0.66	0.77	−0.08	1.7
5	63	EW	0.93	6.7	−8.7	0.77	0.74	−0.06	1.2
5	42	EW	0.88	6.4	−8.8	0.73	0.74	−0.07	1.5
4	21	IV-GJR	0.91	7.3	−10.5	0.69	0.74	−0.07	3.3
4	63	EW (Round 3)	0.92	8.0	−11.5	0.69	0.73	−0.09	1.4

Dollar-neutral, net of 5 bp. Boot.: Calmar across 1,000 stationary-bootstrap resamples of DEV. Turnover: one-way, times per year. ♦: selected.

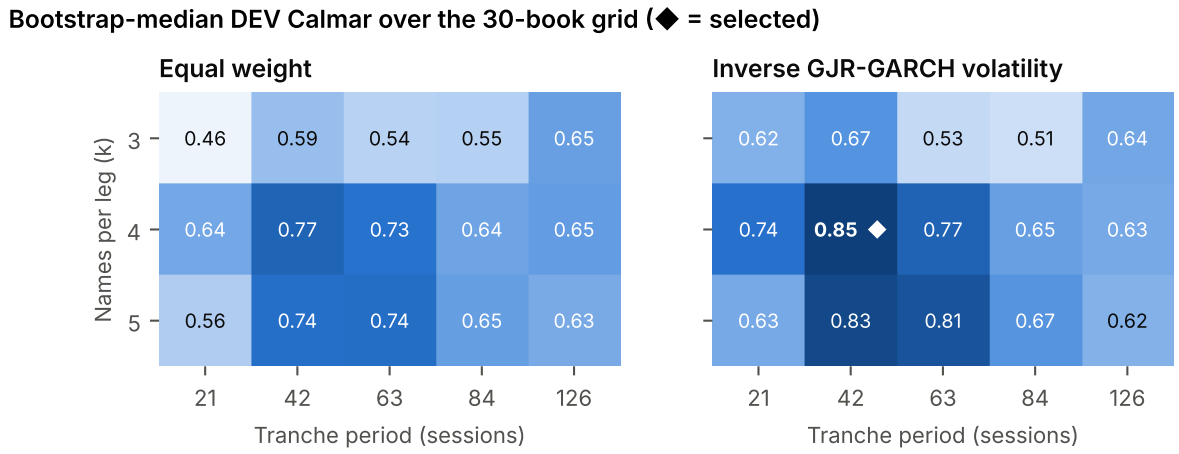


Figure 3. Bootstrap-median development Calmar ratio across the 30-book construction grid.

configurations, and 0.18 against all selection (Table 5). On its own, the development Sharpe ratio is therefore not sufficient evidence, which is why Sections 8 and 9 matter.

7.3 Performance

Table 6 and Figures 4–5 report the frozen book against its predecessors and against momentum. Net of costs, FMF-N earns Sharpe ratios of 0.98, 1.33 and 1.13 in DEV, holdout and full period, with volatility of 9.6%, a maximum drawdown of −14.1%, a beta of −0.02 and a Jensen alpha of 11.9% per year ($t = 2.32$).

- **Against RTB's selection leg** (Calmar 0.36, 0.85 and 0.53, gross of costs), FMF-N roughly

Table 5. Deflated Sharpe ratio of the selected book (development)

Trial set	N_{eff}	SR_0 (ann.)	Sharpe (ann.)	PSR(0)	DSR
Within the 30-book grid	3.5	0.07	0.98	0.95	0.94
Audit research log (71 configurations)	37.2	1.15	0.98	0.95	0.38
All selection incl. pre-log (≈284)	250.2	1.51	0.98	0.95	0.18

SR_0: expected maximum annualised Sharpe ratio of N_{eff} independent null trials. Rows 2–3 use the cross-trial Sharpe dispersion of the research log.

Table 6. Performance of the frozen specification and its predecessors

	CAGR	Vol	Sharpe	MaxDD	Calmar	Beta	α (%)	$t(\alpha)$	PSR(0)
Panel A: DEV, 2017–2019									
Revised strategy ($\tau = 0$, net 5 bp)	7.7	7.9	0.98	−9.7	0.79	−0.01	8.2	1.82	0.95
gross of costs	7.9	7.9	1.01	−9.6	0.82	−0.01	8.4	1.87	0.96
net 10 bp	7.5	7.9	0.95	−9.7	0.77	−0.01	8.0	1.77	0.95
Original selection component (gross)	5.0	8.4	0.63	−13.9	0.36	−0.07	7.5	1.78	0.86
12-1 momentum, same construction (net)	4.8	8.0	0.63	−11.5	0.42	0.07	2.9	0.51	0.86
Original full strategy, HMM tilt (gross)	16.6	10.2	1.56	−9.4	1.77	0.20	10.2	2.46	1.00
Equal-weight market	32.7	18.9	1.59	−25.1	1.31	1.00	–	–	1.00
Panel B: Holdout, 2020–2021									
Revised strategy ($\tau = 0$, net 5 bp)	16.0	11.7	1.33	−14.1	1.14	−0.14	21.1	2.19	0.97
gross of costs	16.3	11.7	1.35	−14.0	1.17	−0.14	21.4	2.22	0.97
net 10 bp	15.7	11.7	1.31	−14.1	1.11	−0.14	20.8	2.16	0.97
Original selection component (gross)	13.2	13.3	1.00	−15.5	0.85	−0.11	17.5	2.07	0.92
12-1 momentum, same construction (net)	5.2	14.2	0.43	−21.8	0.24	−0.17	12.4	1.15	0.72
Original full strategy, HMM tilt (gross)	22.5	14.7	1.45	−13.6	1.65	0.03	20.9	1.99	0.98
Equal-weight market	34.1	32.1	1.08	−38.1	0.89	1.00	–	–	0.93
Panel C: Full period, 2017–2021									
Revised strategy ($\tau = 0$, net 5 bp)	11.0	9.6	1.13	−14.1	0.78	−0.02	11.9	2.32	0.99
gross of costs	11.2	9.6	1.16	−14.0	0.80	−0.02	12.2	2.36	0.99
net 10 bp	10.7	9.6	1.11	−14.1	0.76	−0.02	11.7	2.27	0.99
Original selection component (gross)	8.3	10.6	0.80	−15.5	0.53	−0.07	10.9	2.35	0.96
12-1 momentum, same construction (net)	5.0	10.9	0.50	−21.8	0.23	−0.02	6.4	1.00	0.86
Original full strategy, HMM tilt (gross)	19.0	12.2	1.48	−13.6	1.39	0.11	15.0	2.48	1.00
Equal-weight market	33.3	25.0	1.28	−38.1	0.87	1.00	–	–	1.00

Percent per year for CAGR, volatility, drawdown and alpha. Beta and Jensen alpha from weekly regressions on the equal-weight market with HAC(4) errors. RTB figures are gross of costs.

doubles the development Calmar ratio and improves it in every window.

- **Against 12-1 momentum built identically** (full-period Sharpe 0.50, Calmar 0.23), FMF-N is more than twice as efficient.
- **Costs** are secondary: turnover is 2.1–2.7 times per year, against 10.8 for RTB, and the full-period Sharpe ratio is 1.16 gross and 1.11 at 10 bp.

7.4 Attribution, placebos and concentration

Against the market and four factor books built identically from the price controls, FMF-N loads on 12-1 momentum (0.41) and against reversal (−0.34), which explain 0.60 of its variance, and it retains an alpha of 7.4% per year ($t = 2.98$; Table 7).

- **Placebos** (Table 8, Figure 6). FMF-N beats IID random scores and stock-label permutations in every window (full period $p = 0.003$ and 0.002). Against the circular time-shift it clears the holdout ($p = 0.025$) but not DEV or the full period ($p = 0.076$). Part of the edge is therefore which stocks the signal favours.
- **Leave-one-stock-out** (Table 9, Figure 7). Each rebuild re-runs the entire pipeline on nine stocks. The full-period Sharpe ratio ranges from 0.80 to 1.25 (median 1.06). Removing the largest P&L contributor (stock_6, about half of P&L) leaves a Sharpe of 1.08 and a Calmar of 0.77. No single name carries the result.

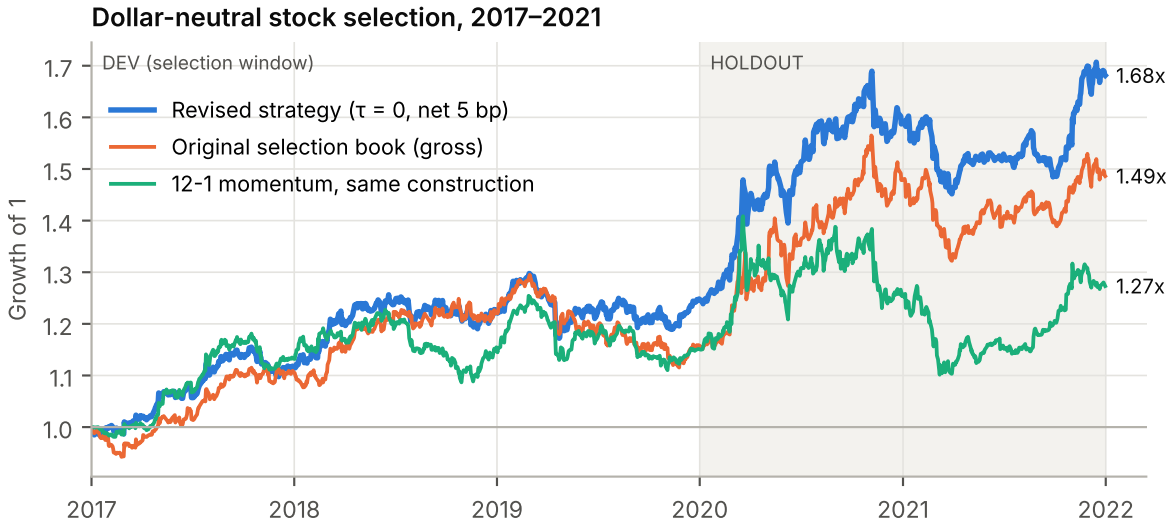


Figure 4. Growth of one unit, 2017–2021. FMF-N and 12-1 momentum are net of 5 bp per unit traded; RTB's selection leg is gross. Shaded: holdout.

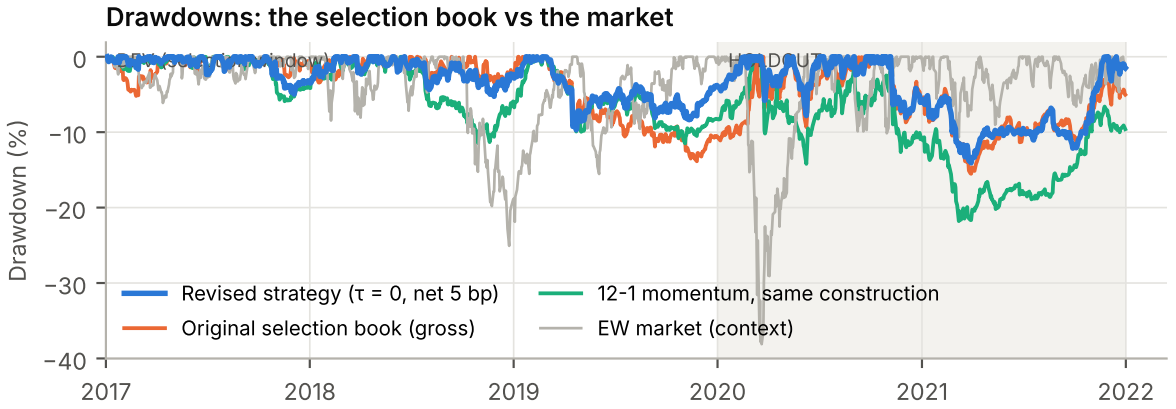


Figure 5. Drawdowns, 2017–2021, with the equal-weight market for context. In February–April 2020 the market's peak-to-trough return was −38.1%, against −5.8% for FMF-N.

Table 7. Factor attribution of the weekly book return

Window	α (%)	$t(\alpha)$	Market	Mom 12-1	Mom 6-1	Rev 1m	Low vol	R^2
DEV	6.0	2.33	−0.03	0.67	0.08	−0.34	0.08	0.67
Holdout	12.3	2.03	−0.05	0.20	0.26	−0.53	−0.07	0.63
Full	7.4	2.98	0.02	0.41	0.25	−0.34	0.05	0.60

OLS on non-overlapping weekly returns, HAC(4) errors, factor books built with the strategy's construction.

Table 8. Placebo tests through the identical construction

Window	Null	Draws	Sharpe	Null 95th	p	Calmar	Null 95th	p
DEV	IID random scores	300	0.98	0.72	0.033	0.79	0.59	0.033
	Stock-label permutation	500	0.98	0.75	0.022	0.79	0.60	0.026
	Circular time-shift	315	0.98	1.39	0.146	0.79	1.52	0.146
Holdout	IID random scores	300	1.33	0.80	0.010	1.14	0.69	0.017
	Stock-label permutation	500	1.33	0.98	0.016	1.14	1.06	0.044
	Circular time-shift	315	1.33	1.24	0.025	1.14	1.19	0.060
Full	IID random scores	300	1.13	0.44	0.003	0.78	0.22	0.003
	Stock-label permutation	500	1.13	0.56	0.002	0.78	0.37	0.004
	Circular time-shift	315	1.13	1.21	0.076	0.78	1.05	0.076

$$p = (1 + \#\{\text{null} \geq \text{observed}\})/(1 + \text{draws}).$$

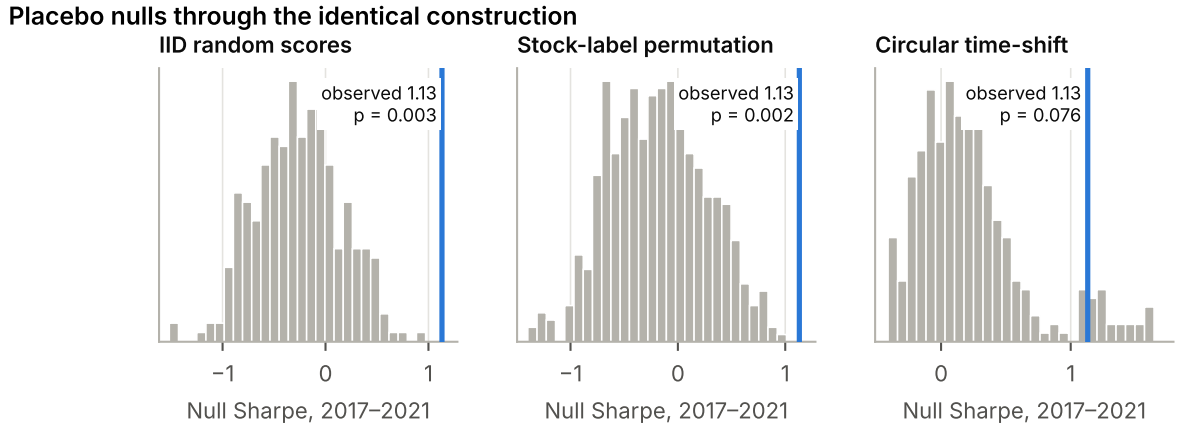


Figure 6. Full-period Sharpe ratios under the three placebo nulls, the line marks the observed value.

Table 9. Full leave-one-stock-out rebuilds

	Sharpe			Calmar			MaxDD	α	$t(\alpha)$
Universe	DEV	Hold.	Full	DEV	Hold.	Full	Full (%)	Full (%)	Full
All ten stocks	0.98	1.33	1.13	0.79	1.14	0.78	−14.1	11.9	2.32
without stock_1	1.14	1.14	1.12	1.00	0.92	0.75	−14.7	12.4	2.39
without stock_2	0.58	1.10	0.81	0.47	0.96	0.62	−10.7	5.5	1.37
without stock_3	0.78	0.91	0.83	0.63	0.65	0.49	−14.3	9.0	2.03
without stock_4	0.90	0.72	0.80	0.77	0.50	0.46	−15.8	8.6	1.72
without stock_5	1.03	1.53	1.25	0.78	1.56	1.05	−11.4	13.3	2.73
without stock_6	0.87	1.34	1.08	0.66	1.18	0.77	−13.8	10.6	1.98
without stock_7	1.03	1.32	1.15	0.80	1.17	0.80	−14.1	12.9	2.42
without stock_8	0.80	1.27	1.02	0.63	1.16	0.73	−12.8	10.4	1.99
without stock_9	0.86	1.47	1.14	0.66	1.39	0.84	−12.5	11.1	2.33
without stock_10	0.98	1.15	1.04	1.23	0.91	0.65	−14.7	10.3	2.13

Each row re-runs cleaning, calendar, snapshot, features, cross-sectional ranks, composite and allocation without one stock. GARCH forecasts are univariate and therefore unchanged.

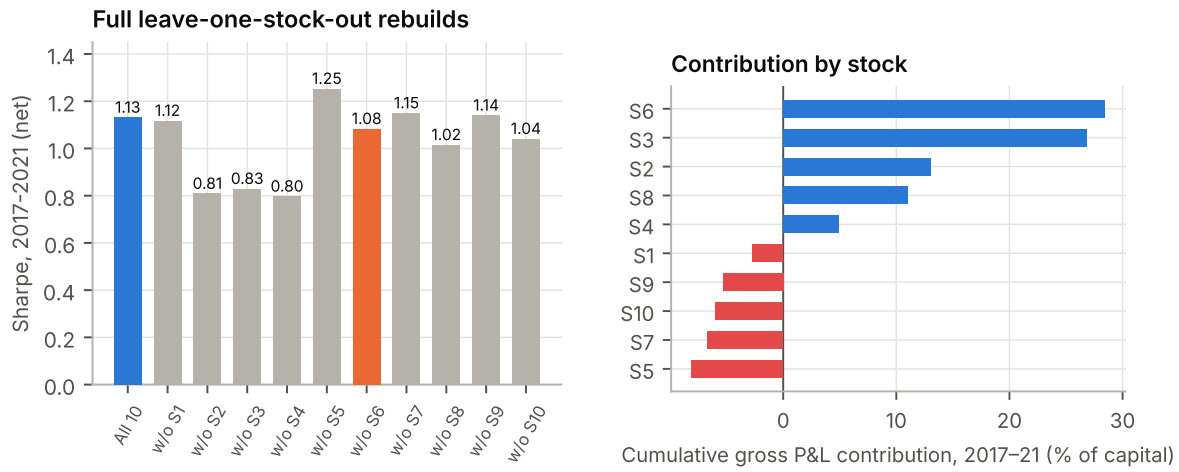


Figure 7. Left: full-period Sharpe ratio of the ten leave-one-stock-out rebuilds (orange: without the largest contributor). Right: cumulative gross P&L by stock.

8 First-Mover Flow versus the institutional toolkit

A selection signal is valuable to a portfolio manager only if it adds something the standard toolkit does not. I build 13 cross-sectional comparables (Appendix B) through FMF-N's identical construction, so that differences isolate the signal, and add three allocations. The comparables cover the price factors, the sell-side signals in common use, and two multi-factor blends.

League table: Among the 14 cross-sectional books, FMF-N has the highest full-period Sharpe (1.13) and Calmar (0.78) ratios (Table 10, Figure 8). The nearest are classic revision breadth (0.92 and 0.58), one of FMF-N's components, and the momentum + revisions blend (0.87 and 0.49). Plain 12-1 momentum reaches 0.50 and 0.23. Betting-against-beta, low dispersion and reversal lose money in this universe.

Where the toolkit fails: Figure 9 and Table 11 report five named episodes.

- **COVID crash:** The market returned -35.5%, betting-against-beta -15.2% and reversal -7.7%, while FMF-N returned $+12.0\%$.
- **Vaccine rotation (November 2020):** In this momentum crash FMF-N returned -5.8%, against -8.2% for 12-1 and -8.2% for residual momentum.
- **Feb-Mar 2021 growth sell-off:** FMF-N's worst episode (-7.1%) was milder than 12-1 momentum (-10.5%) and time-series momentum (-12.1%).

Spanning: Regressed on the market and *all* 13 comparables, FMF-N's alpha shrinks to 1.8% ($t = 1.46$, $R^2 = 0.92$; Table 12). That test is circular, because two comparables contain FMF-N's components. Against the market and the 11 comparables *not* built from its components, the alpha is 6.1% per year ($t = 2.51$) over 2017–2021 and 7.5% ($t = 3.86$) in DEV.

Table 10. FMF-N and institutional comparables, identical construction (net of 5 bp)

	Sharpe			Calmar			MaxDD	Beta	Corr.
	DEV	Hold.	Full	DEV	Hold.	Full	Full (%)	Full	FMF-N
FMF-N (this paper)	0.98	1.33	1.13	0.79	1.14	0.78	−14.1	−0.02	1.00
Revision breadth (classic)	0.57	1.28	0.92	0.38	1.08	0.58	−15.5	−0.02	0.95
Momentum + revisions blend	0.78	1.01	0.87	0.52	0.73	0.49	−18.5	−0.01	0.90
Multi-factor: mom + low vol + reversal	0.88	0.80	0.81	0.71	0.52	0.42	−14.3	−0.09	0.44
Low volatility (60d)	0.37	0.80	0.56	0.21	1.18	0.37	−13.6	−0.10	0.04
12-1 momentum (Jegadeesh-Titman)	0.63	0.43	0.50	0.42	0.24	0.23	−21.8	−0.02	0.71
Target-price upside (consensus)	0.47	0.29	0.38	0.37	0.24	0.24	−12.1	0.04	−0.38
52-week-high proximity	−0.36	1.06	0.37	−0.16	0.75	0.17	−18.2	−0.09	0.73
Target-price momentum (63d)	−0.36	0.68	0.19	−0.15	0.44	0.07	−20.4	−0.08	0.71
Residual momentum (Blitz et al.)	0.42	0.00	0.17	0.24	−0.04	0.05	−24.1	−0.06	0.64
6-1 momentum	−0.51	0.55	0.06	−0.16	0.32	0.00	−24.8	−0.04	0.67
Betting-against-beta (low beta)	−0.28	0.00	−0.15	−0.17	−0.05	−0.07	−32.2	0.01	0.10
Short-term reversal (1m)	0.25	−0.69	−0.24	0.16	−0.33	−0.09	−18.7	0.04	−0.51
Low analyst dispersion	−0.36	−0.74	−0.53	−0.19	−0.50	−0.20	−27.7	−0.13	−0.15
Allocation benchmarks									
Equal-weight long-only (market)	1.59	1.08	1.28	1.31	0.89	0.87	−38.1	1.00	−0.02
Inverse-vol long-only (naive risk parity)	1.67	1.17	1.35	1.37	0.99	0.92	−37.2	0.96	−0.01
Time-series momentum 12-1 (CTA-style)	0.89	0.86	0.86	0.57	0.89	0.54	−30.2	0.61	0.29
RTB: original regime-timed breadth (gross)	1.56	1.45	1.48	1.77	1.65	1.39	−13.6	0.11	0.63
FMF-B20: FMF + explicit 0.2 beta budget	1.56	1.73	1.62	1.82	2.17	1.66	−10.8	0.17	0.89

Cross-sectional comparables: top/bottom four, inverse-GJR weights, 42-session tranches, $\tau = 0$, net of 5 bp. Allocation benchmarks in their natural form. RTB gross, FMF-B20 adds a constant 0.2 beta budget to FMF-N.

Table 11. Named episodes: cumulative returns (%)

	Q4-2018 10/03–12/24/18	COVID crash 02/19–03/23/20	COVID rebound 03/24–06/08/20	Vaccine rotation 11/06–11/30/20	Growth sell-off 02/12–03/08/21
FMF-N	−0.8	12.0	−4.0	−5.8	−7.1
12-1 momentum (Jegadeesh-Titman)	−0.6	12.2	−9.8	−8.2	−10.5
Residual momentum (Blitz et al.)	5.3	8.3	−11.1	−8.2	−10.4
Low volatility (60d)	2.6	−0.6	2.8	2.8	1.2
Betting-against-beta (low beta)	1.3	−15.2	3.0	−1.0	−1.7
Short-term reversal (1m)	1.9	−7.7	2.4	4.0	1.1
Target-price upside (consensus)	3.1	−3.7	6.3	7.3	1.7
Revision breadth (classic)	−1.7	10.2	−2.3	−7.4	−6.1
Multi-factor: mom + low vol + reversal	4.2	6.0	−5.8	−0.7	−4.7
Time-series momentum 12-1 (CTA-style)	−23.3	−18.9	18.5	−3.5	−12.1
Equal-weight long-only (market)	−24.8	−35.5	60.9	6.9	−9.4
RTB: original regime-timed breadth	−4.1	11.9	−1.1	−5.5	−9.2

Enhancement: A 50/50 blend with FMF-N raises the Sharpe ratio of 16 of 18 comparables and the Calmar ratio of 16 of 18 (Figure 10). The two exceptions are the books that already carry market beta (RTB and FMF-B20), whose Calmar ratios are driven by the bull market.

Construction is matched, not imposed: Slow tranches help the slow analyst-flow signals: classic breadth has a full-period Sharpe of 0.92 in tranches against 0.52 in its native monthly form. They do not help every price signal: 12-1 momentum has 0.74 native against 0.50 in tranches.

9 Out of sample: a never-seen period

Protocol: Before any return on the new universe was computed, the transfer rules were written to a file and hashed (Appendix C). The only information examined beforehand was structural: the file layout, the date range, rating counts, the cleaning log and signal coverage.

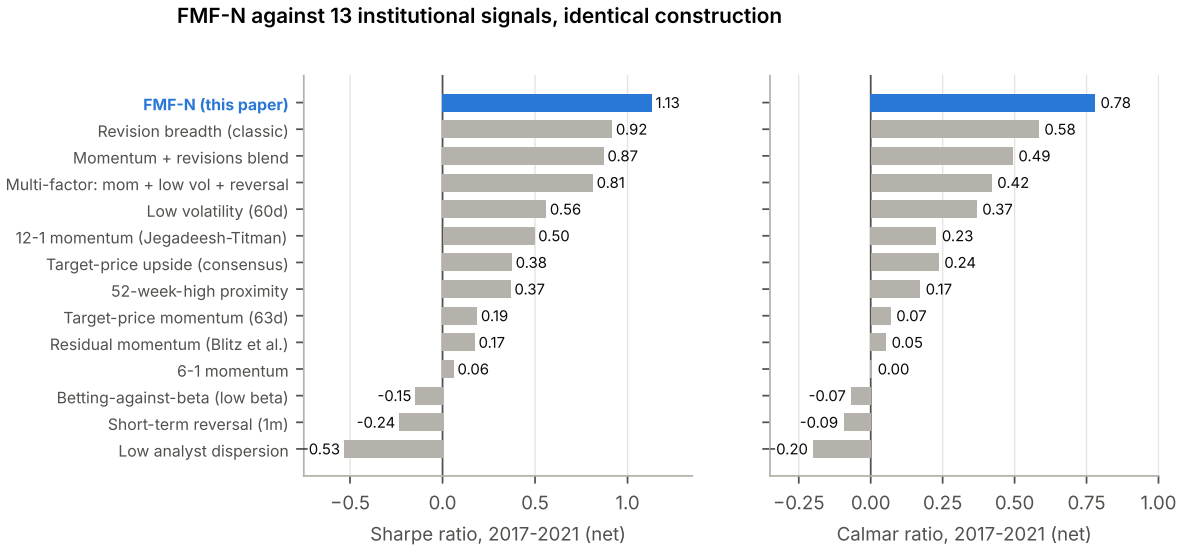


Figure 8. Full-period Sharpe and Calmar ratios of FMF-N and 13 institutional signals built through the identical construction.

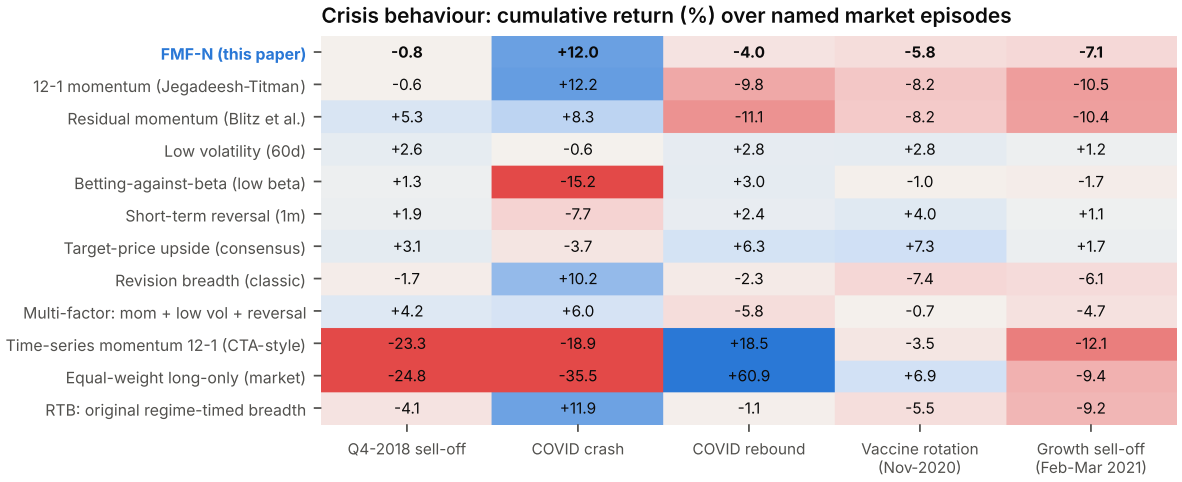


Figure 9. Cumulative return over named market episodes (blue: gain, red: loss).

Table 12. Spanning regressions of the weekly FMF-N return

Regressors	Window	α (% p.a.)	$t(\alpha)$	R^2
Market + all 13 comparables	DEV	2.4	2.19	0.95
	Holdout	1.2	0.50	0.93
	Full	1.8	1.46	0.92
Market + 11 comparables not built from FMF components	DEV	7.5	3.86	0.77
	Holdout	6.4	1.02	0.69
	Full	6.1	2.51	0.69

OLS on non-overlapping weekly returns, HAC(4) errors. The first panel includes classic breadth and the momentum + revisions blend, both built from FMF-N's components.

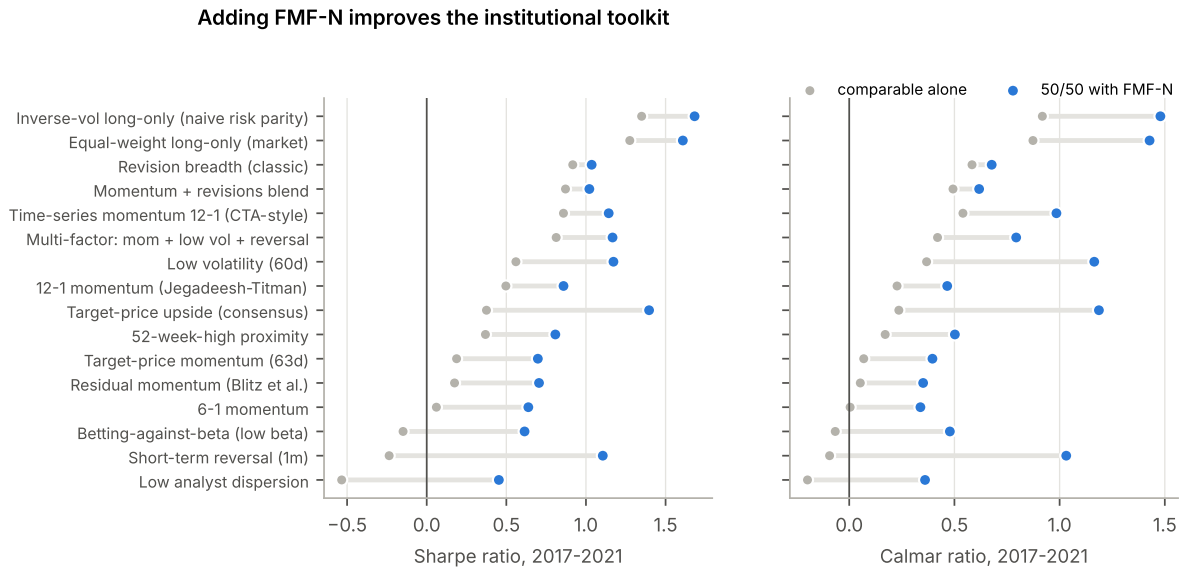


Figure 10. Full-period Sharpe and Calmar ratios of each comparable alone (grey) and blended 50/50 with FMF-N (blue).

- **Frozen from development:** signals, LMH parameters, tranche period, allocator, costs and data rules.
- **Names per leg:** the development choice of 4 names out of 10 transfers as a *leg fraction* of 0.4 of the names with a valid score each session, $k_t = \max(2, \lfloor 0.4 N_t \rfloor)$ (primary). The literal $k = 4$ is the secondary rule.
- **Evaluation window:** starts after a 252-session warm-up and runs from 2024-01-05 to 2026-08-05 (647 sessions).

Nothing was changed afterwards.

Results: The primary book earns a net Sharpe ratio of 1.05, a CAGR of 10.5% at 10.0% volatility, a maximum drawdown of −8.0% and a Calmar ratio of 1.32 (Table 13, Figure 11). Its beta is −0.02 and its Jensen alpha is 11.4% per year ($t = 2.04$). The PSR is 0.95, deflating over the two pre-registered variants leaves it unchanged, because they are highly correlated. Both halves of the window are positive (Sharpe 1.44 and 0.88). The literal $k = 4$ rule, with four names per leg out of about 19, is weaker (Sharpe 0.70, Calmar 0.72): in a wider universe, breadth should scale with the universe, as pre-registered.

The novel signal replicates: LMH has an IC21 of 0.096 (one-sided $p = 0.042$) out of sample, essentially the development value (0.095, Table 14, Figure 12). Breadth (0.051) and the composite (0.073, $p = 0.065$) are positive. Event breadth is negative but covers only 26% of stock-days, because clusters of five same-day ratings are rare with 13 analysts per stock.

Against the toolkit, out of sample: FMF-N ranks 2nd of 14 cross-sectional books on both Sharpe and Calmar (Table 16, Figure 12). Only target-price momentum is ahead in this period (1.14 and 1.44). A 50/50 blend with FMF-N raises the Sharpe and Calmar ratios of all 16

Table 13. Out of sample: frozen FMF-N over the never-seen period (net of 5 bp)

	CAGR	Vol	Sharpe	MaxDD	Calmar	Beta	α	$t(\alpha)$	PSR	Sharpe H1	Sharpe H2
FMF-N, primary (leg fraction 0.4)	10.5	10.0	1.05	−8.0	1.32	−0.02	11.4	2.04	0.95	1.44	0.88
FMF-N, secondary ($k = 4$)	7.3	10.9	0.70	−10.2	0.72	−0.04	8.8	1.48	0.87	0.94	0.64
FMF-B20, beta budget 0.2 (exploratory)†	15.9	10.3	1.49	−6.5	2.43	0.16	11.8	2.09	0.99	1.73	1.44
Equal-weight market (30 stocks)	23.6	20.4	1.14	−25.3	0.94	1.00	–	–	0.97	0.28	2.35

Evaluation window 2024-01-05 to 2026-08-05. Percent per year for CAGR, volatility, drawdown and alpha. H1 and H2: halves of the window. †Not in the pre-registration, the budget was fixed in development. Reported for information and used in no test.

comparables. The market and CTA-style trend have high Sharpe ratios in this bull period, but with betas of 1.00 and 0.43 and drawdowns of −25.3% and −10.8%.

Placebos and attribution: The book beats IID random scores ($p = 0.027$) and is borderline against stock-label permutation ($p = 0.056$). Against the circular time-shift it is not significant ($p = 0.226$), the same pattern as in development. After the market and the four price-control books, the alpha is 7.3% ($t = 1.79$, $R^2 = 0.47$). The window is 2.6 years, so this is one more sample rather than proof. It is, however, the first sample that was never used for any choice, and the method's two signatures, zero beta with positive alpha and LMH predicting returns, both reappear in it.

Is the result carried by the ten development names? Ten of the thirty names are the development stocks, observed again two years later. A post-hoc split, not part of the pre-registration, re-runs the frozen specification on each part separately with the same GARCH forecasts. On the twenty names never used in development the book earns a net Sharpe ratio of 0.80, a Calmar ratio of 0.98 and a maximum drawdown of −9.2%, and LMH keeps its information coefficient (IC21 0.105, one-sided $p = 0.058$). The ten development names alone earn 0.31, with LMH near zero (0.022), but with sparse broker coverage a valid cross-section exists on only part of the window. The out-of-sample result is therefore not a re-run of known names; if anything, the new names carry it. The selection of the twenty names is a different matter: they are today's large caps, a choice that knows which companies grew large. A dollar-neutral book is exposed to that choice only through the spread between its legs, and the static tilt towards particular names loses money out of sample (Section 10.1).

An exploratory check of the beta budget: The pre-registration lists only the two market-neutral variants, so the beta-budget variant is reported separately and enters no test. With the same scores and leg fraction and the budget $c = 0.2$ fixed in development, FMF-B20 earns a Sharpe ratio of 1.49, a Calmar ratio of 2.43 and a maximum drawdown of −6.5% at a beta of 0.16. Its Jensen alpha is unchanged (11.8% a year, $t = 2.09$), as it should be: the budget adds market return, not skill. In a bull window this is the expected direction. In a falling market the budget costs about c times the market's loss, which is why it is a disclosed constant rather than a timed bet.

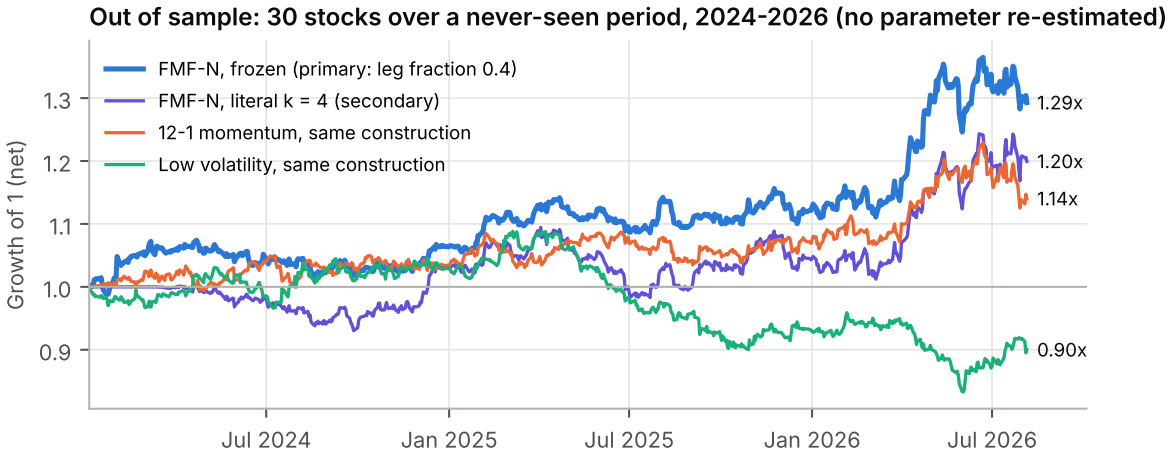


Figure 11. Growth of one unit over the never-seen period, frozen specification, net of 5 bp.

Table 14. Out-of-sample signal evidence (labels inside the evaluation window)

	IC21	t_{phase}	p	Hit (%)	Coverage (%)
breadth_63	0.051	0.98	0.168	58	47
event_breadth_63	−0.041	−0.49	0.685	46	26
breadth_resid_mom	0.020	0.39	0.351	52	46
leader_minus_herd	0.096	1.80	0.042	64	53
Composite	0.073	1.56	0.065	58	59
leaders only	0.087	1.70	0.050	61	74

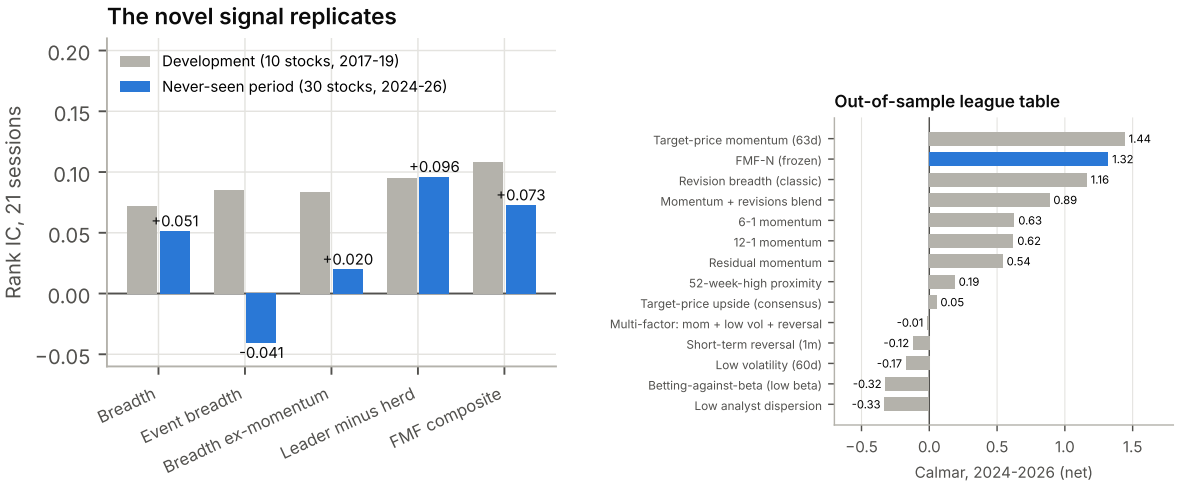


Figure 12. Left: rank IC (21 sessions) of each signal in development and over the never-seen period. Right: out-of-sample Calmar ratios of FMF-N and the cross-sectional comparables.

Table 15. Out-of-sample placebo tests

Null	Draws	Sharpe	Null median	Null 95th	p
IID random scores	300	1.05	−0.26	0.87	0.027
Stock-label permutation	500	1.05	−0.01	1.07	0.056
Circular time-shift	216	1.05	0.09	2.17	0.226

Table 16. Out of sample: FMF-N and institutional comparables

	Sharpe	Calmar	MaxDD (%)	Beta	Corr. FMF-N	50/50 with FMF-N	
						Sharpe	Calmar
Time-series momentum 12-1 (CTA-style)	1.30	1.62	−10.8	0.43	0.33	1.45	1.86
Inverse-vol long-only (naive risk parity)	1.23	1.01	−22.5	0.88	−0.10	1.66	1.56
Equal-weight long-only (market)	1.14	0.94	−25.3	1.00	−0.09	1.54	1.41
Target-price momentum (63d)	1.14	1.44	−7.5	0.04	0.67	1.19	1.76
FMF-N (frozen)	1.05	1.32	−8.0	−0.02	1.00	–	–
Revision breadth (classic)	0.98	1.16	−8.7	−0.01	0.87	1.05	1.24
Momentum + revisions blend	0.79	0.89	−8.2	0.08	0.66	1.01	1.26
12-1 momentum (Jegadeesh-Titman)	0.60	0.62	−8.5	0.09	0.61	0.92	1.10
6-1 momentum	0.56	0.63	−7.2	0.04	0.63	0.91	1.09
Residual momentum (Blitz et al.)	0.56	0.54	−7.9	0.02	0.66	0.91	1.08
52-week-high proximity	0.31	0.19	−12.0	−0.09	0.66	0.79	0.81
Target-price upside (consensus)	0.11	0.05	−11.7	0.05	−0.47	1.19	1.50
Multi-factor: mom + low vol + reversal	0.00	−0.01	−14.3	−0.13	0.39	0.77	0.71
Short-term reversal (1m)	−0.14	−0.12	−7.9	0.00	−0.41	1.05	1.07
Low volatility (60d)	−0.39	−0.17	−23.5	−0.26	0.12	0.47	0.31
Low analyst dispersion	−0.89	−0.33	−25.5	−0.28	−0.13	0.17	0.10
Betting-against-beta (low beta)	−0.94	−0.32	−34.3	−0.33	−0.19	−0.04	−0.04

10 Attacking the result: nine objections and what the data say

A positive backtest in a ten-stock universe should be met with suspicion, and the suspicion is professional rather than personal. This section states the objections that a quantitative researcher, a portfolio manager and a QIS structurer would raise, and confronts each with a test designed to break the result. Table 17 summarises the outcome. The tests are run on the frozen specification and, where an objection concerns the research process rather than the strategy, on the entire family of specifications a researcher could reasonably have chosen. Every test is reproducible from the repository (`scripts/reproduce_audit.py`) and the notebook (Block 17).

10.1 Identity or timing?

The most serious objection to any backtest on a named universe is hindsight. The ten development stocks are well-known large caps whose 2016–2021 paths were known when the universe was assembled. A book that is *on average* long the names that later did well earns money without any skill in timing, and the circular time-shift placebo of Section 7, which keeps each stock's habitual rank, cannot rule this out. Lo (2008) provides the decomposition. The mean return of a book with weights $w_{i,t}$ is

$$E\Big[\sum_i w_{i,t-1} r_{i,t}\Big] = \underbrace{\sum_i \mathrm{Cov}(w_{i,t-1}, r_{i,t})}_{\text{active: timing}} + \underbrace{\sum_i E[w_{i,t-1}]\,E[r_{i,t}]}_{\text{passive: identity}}. \tag{6}$$

Because the passive term uses in-window averages, it is a diagnostic. I therefore also construct two *tradable* books: a static book holding each stock's expanding mean of past weights, $\bar{w}_{i,t} = \frac{1}{t}\sum_{s\le t} w_{i,s}$, and a timing book holding $w_{i,t} - \bar{w}_{i,t}$. They sum to the strategy and use no future information.

The answer is unambiguous (Figure 13). In the development universe the timing book earns 8.9% a year with a Sharpe ratio of 1.04 ($t_{\mathrm{HAC}} = 2.41$), while the static book earns 2.2%

Table 17. Anticipated objections, the test applied and the verdict

Objection	Test	Result
1. The universe was chosen with hindsight, the book is long the later winners.	Lo (2008) decomposition, causal static and timing books, partial pooling of per-stock P&L	The timing book carries the edge (Sharpe 1.04, $t = 2.41$, never-seen 1.33, $t = 2.21$). The static tilt is insignificant and negative out of sample.
2. Execution at the signal close and free shorting are unrealistic.	0–2 session delay × 0–200 bp borrow × 0–30 bp costs, breakeven cost	Sharpe 0.96 (0.89 never-seen) in the harshest cell, breakeven cost 234 bp per unit traded.
3. Overlapping tranches smooth P&L and inflate the Sharpe ratio.	Lo (2002) autocorrelation-adjusted Sharpe	1.06 and 1.06, first-order autocorrelation 0.04.
4. Zero average beta hides crash beta.	Up- and down-market betas, worst market weeks	Down-market beta -0.07 (t =-1.24), positive mean return in the worst market decile.
5. The parameters were chosen to make it work.	Multiverse of ≈300 specifications, SPA, PBO, selection-free ensemble	99% (97%) of specifications earn a positive Sharpe, the ensemble earns 0.92 ($t = 1.97$) and 0.86.
6. Beating the toolkit is cherry-picking.	Romano–Wolf step-down against 16 books, reverse SPA	No comparable beats FMF-N (SPA $p = 0.86$, 0.86), FMF-N significantly beats 2 books after FWER control in sample.
7. The new signal adds nothing to classic breadth.	FMF-N vs the same book without LMH, pooled across samples	$+0.28$ Sharpe ($p = 0.028$) in sample, $+0.30$ out of sample, $P(\text{gain} > 0) = 0.94$ pooled.
8. Ten stocks and a few years prove nothing.	Bayesian Sharpe under sceptical priors, random-effects meta-analysis, empirical-Bayes shrinkage, Bayes factors, power	$P(\text{IC} > 0 \text{ in the next universe}) = 0.96$, pooled IC 0.105, Stouffer $p = 0.0022$, the 18-signal search shrinks LMH's t to 1.06.
9. One history has one drawdown.	Generative stress tests (bootstrap, FHS, Diffusion-FHS, TimeGAN) with posterior-predictive drift	Five-year median Calmar 0.70, 5% max drawdown -34%, the binding stress is a momentum crash (5% three-month quantile about -15%), not a market crash.

Development: 2017–2021, 10 US large caps. Never-seen period: 2024-01-05 to 2026-08-05, 30 stocks (20 never used in development). Details in Sections 10.1–11.

($t = 1.08$). Over the never-seen period the contrast sharpens: timing earns a Sharpe ratio of 1.33 ($t = 2.21$) and the static tilt *loses* money (Sharpe -0.49). The edge is *when* the revision flow turns, not *which* stocks it habitually favours. The time-shift placebo is therefore best read as a low-power test: its null distribution is wide because shifted signals keep their slow-moving structure, not because identity explains the returns.

A Bayesian partial-pooling model of per-stock P&L contributions (normal–normal, with the cross-stock heterogeneity estimated) confirms that the development result is not one name. The largest contributor (stock_6, 5.7% a year) is shrunk to 4.5%, and the posterior probability that the typical stock contributes positively is 0.87. Out of sample, the contributions are small and diffuse across the 30 names. Ten of those names are the development stocks; on the twenty others alone the frozen book earns a Sharpe ratio of 0.80 and LMH an IC of 0.105 (Section 9), so the out-of-sample timing edge does not come from names already studied.

10.2 Implementation: delay, borrow, costs, smoothing and crash beta

Four practical objections have to be raised here to be sure that this does not turn a paper alpha into a live loss. Table 17 and Figure 14 report the full grid.

- **Execution delay:** Trading one session after the signal close leaves the Sharpe ratio at 1.13 (development) because the book is re-targeted by one forty-second each day. Information that builds over weeks is not front-run by a day.
- **Borrow and costs:** With a one-session delay, 10 bp per unit traded and a 100 bp annual borrow fee on all short notional, the Sharpe ratio is 1.06 (0.99 over the never-seen period). In the harshest cell of the grid (two sessions, 200 bp borrow, 20 bp costs) it is 0.96 and 0.89. The breakeven cost per unit traded is 234 bp (197 bp). This is the cleanest capacity

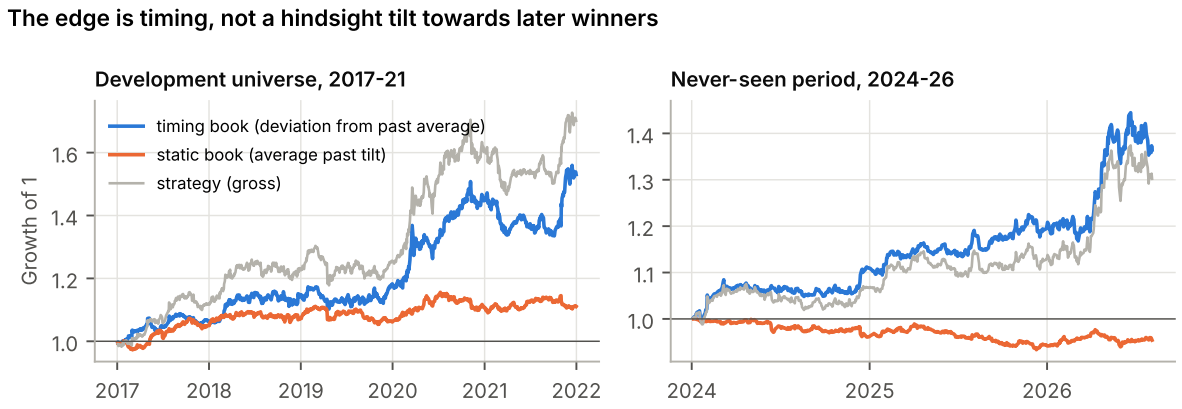


Figure 13. Causal decomposition of the gross book into a static book (each stock's expanding average of past weights) and a timing book (today's deviation from it). Both are tradable and sum to the strategy.



Figure 14. Net Sharpe ratio against trading cost for three implementation regimes: trading at the signal close without borrow fees; one session late with a 100 bp borrow fee; two sessions late with a 200 bp borrow fee.

diagnostic available without volume data, and it is an order of magnitude above the cost of trading liquid large caps.

- **Smoothing:** Overlapping tranches smooth *positions*, not returns. The first-order autocorrelation of daily P&L is 0.04, and the Lo (2002) autocorrelation-adjusted Sharpe ratio is 1.06 (1.06).
- **Crash beta:** Weekly regressions with separate up- and down-market slopes give a down-market beta of -0.07 (t =-1.24) and an up-market beta of 0.03. The book earns a positive average return in the worst decile of market weeks.

10.3 Research degrees of freedom

Every backtest embeds choices: windows, thresholds, legs, tranches, allocators, signal sets, data rules. A result that survives only one combination is an artefact of the search. Following the multiverse analysis of Steegen et al. (2016) and the specification curve of Simonsohn, Simmons, and Nelson (2020), I re-run the entire pipeline over every choice in this research space. The grid includes the leader gap (5–20), the herd window (2–10 to 5–20), the LMH

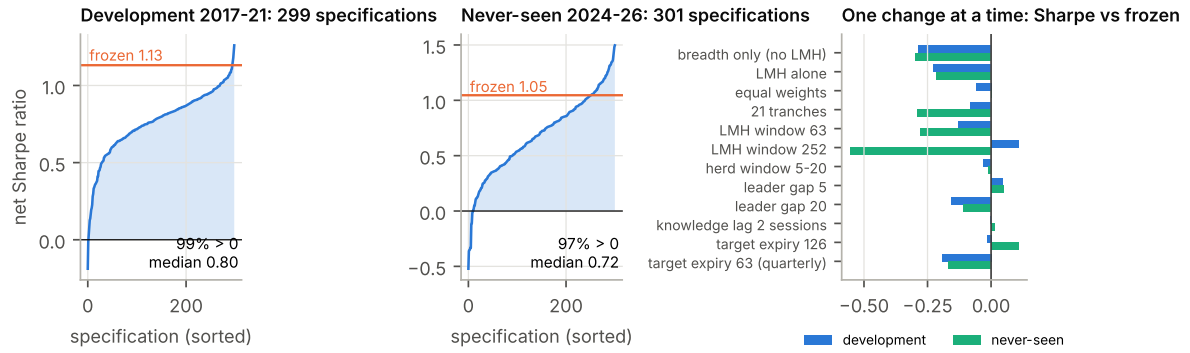


Figure 15. Left and centre: specification curves, the net Sharpe ratio of every randomly drawn specification sorted from worst to best, with the frozen specification marked. Right: change in Sharpe ratio when one choice at a time departs from the frozen specification.

window (63–252), the breadth window, tranche length, legs, allocator, four signal sets, the information-event threshold (4–6), a two-session knowledge lag, and the target expiry. The expiry choices include the *quarterly* 63-session horizon that matches the earnings cycle, as well as 126 and 252 sessions. Each parameter is varied one at a time, and 300 joint specifications are drawn at random.

Three results stand out (Figure 15).

- **The sign is not a choice:** 99% of the development specifications and 97% of the never-seen ones earn a positive net Sharpe ratio, with medians of 0.80 and 0.72. The frozen specification sits at the 98th percentile in development, as selection implies, and at the 83th out of sample, which selection does not imply.
- **The level is:** The family's ranking does not persist (PBO 0.76). The *ensemble* that holds all specifications at once, a selection-free estimate, earns a Sharpe ratio of 0.92 over 2017–2021 ($t = 1.97$) and 0.86 out of sample. That is roughly 0.8 of the frozen specification's performance.
- **Quarterly calibration and knowledge lag:** A hard 63-session expiry lowers the development Sharpe ratio to 0.94 (0.88 out of sample), because dropping a target after one quarter discards the information in analysts who update semi-annually. A 126-session expiry is equivalent to the 252-session default (1.12, 1.15). A two-session knowledge lag changes nothing (1.13, 1.06). Staleness is better handled *inside* the signal, through the event and silence clocks of LMH, than by truncating the snapshot.

The family-wise test is sobering in the right way. On any single short window, Hansen's (2005) SPA test cannot reject that no specification beats zero (development $p = 0.34$, never-seen $p = 0.14$; the ensemble over 2017–2021 has $p = 0.09$). One window is not enough, the evidence comes from combining independent samples, which Section 10.5 does formally.

10.4 Multiple testing against the toolkit, and the marginal value of LMH

Ranking first among 14 cross-sectional books (Section 8) invites the charge of cherry-picking the comparison set. Two family-wise tests answer it. The reverse SPA test asks whether *any* of the 16 institutional books significantly beats FMF-N once each is scaled to the same

volatility. It does not ($p = 0.86$ in sample, 0.86 out of sample). The Romano and Wolf (2005) step-down procedure asks which books FMF-N beats with family-wise error control. In sample, it beats 2 of 16 books, including target-price momentum and 6–1 momentum. Out of sample, none of the differences survives the correction, although 2 are individually significant. The defensible claim is therefore "never beaten, and better than most", not "significantly better than everything".

The sharper question is whether the new signal earns its place. Classic revision breadth is FMF's closest relative (correlation of the books 0.95). I compare FMF-N with the identical book built from the three breadth signals alone, using a circular block bootstrap of the paired returns (Ledoit and Wolf, 2008). The correlation between the two books is what gives the test its power. Adding LMH raises the Sharpe ratio by 0.28 in development (95% interval 0.03 to 0.54, $p = 0.028$) and by 0.30 over the never-seen period (-0.21 to 0.80). Pooled, the gain is 0.29 with a posterior probability of 0.94 that it is positive.

10.5 Small samples, done properly

Frequentist p-values answer a narrow question: how surprising the data are if the effect is exactly zero. With small samples they also discard information that every allocator uses, namely that most strategies do not work, and that the same mechanism has now been observed in three independent samples. I use four Bayesian and small-sample tools (Figure 16).

Posterior Sharpe ratios under sceptical priors: I use the Bayesian bootstrap (Rubin, 1981) of weekly returns, a nonparametric posterior that lets skewness and fat tails enter through the data, re-weighted by a prior on the annual Sharpe ratio. Under a sceptical $\mathrm{N}(0, 1)$ prior, which says that most strategies have $|SR| < 2$, the 2017–2021 posterior mean is 1.00 with $P(SR > 0) = 0.99$. Over the never-seen period it is 0.76 with $P(SR > 0) = 0.93$. Under a very sceptical $\mathrm{N}(0, 0.5)$ prior, which encodes the post-publication decay of anomalies documented by McLean and Pontiff (2016), the probabilities are 0.97 and 0.86.

Random-effects meta-analysis across independent samples: The development period, the holdout and the never-seen period are three non-overlapping samples of the same mechanism (the third adds twenty new names to the ten development names). A normal–normal random-effects model with a half-Cauchy prior on the between-sample heterogeneity τ (Gelman, 2006) pools them *without* assuming that they share one true effect. For LMH, the information coefficients are 0.095, 0.129 and 0.096. The pooled mean is 0.105 (90% HDI 0.033 to 0.179), and the posterior-predictive probability that the IC is positive in the *next* universe is 0.96. Stouffer's combination of the three one-sided p-values gives $p = 0.0022$, and the product of the three Bayes factors (prior sd 0.1) is 22. For the strategy's Sharpe ratio, the pooled mean is 1.10 (90% HDI 0.36 to 1.84), with $P(SR > 0 \text{ in the next universe}) = 0.96$ and $P(SR > 0.5) = 0.86$.

Shrinking the search: LMH was the best of 18 candidate signals in development (Fama–MacBeth $t = 2.46$). Empirical-Bayes (James–Stein) shrinkage of the 18 t-statistics towards their family (Efron, 2010) reduces this to 1.06, with a posterior probability of a positive effect of 0.95. This is the honest value of the development search *taken alone*: suggestive, not conclusive.

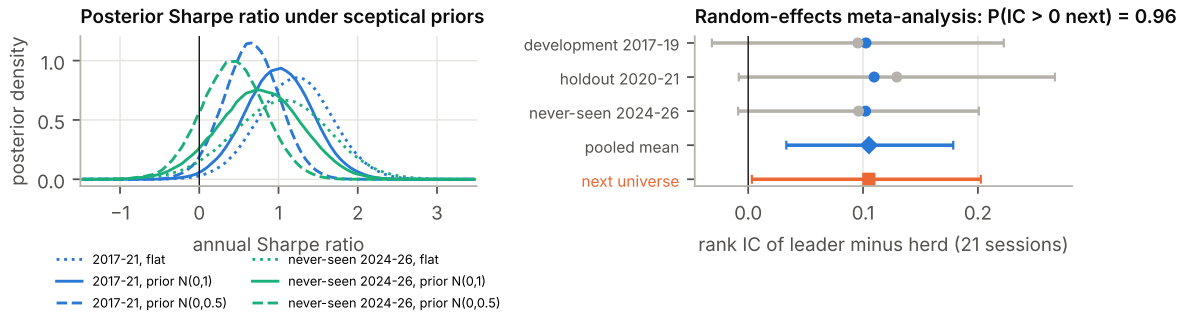


Figure 16. Left: posterior distributions of the annual Sharpe ratio (Bayesian bootstrap of weekly returns) under a flat prior and two sceptical priors. Right: random-effects meta-analysis of the leader-minus-herd IC. Grey: sample estimates with 95% intervals; blue dots: partially pooled estimates; diamond: pooled mean (90% HDI); square: predictive IC in a new universe.

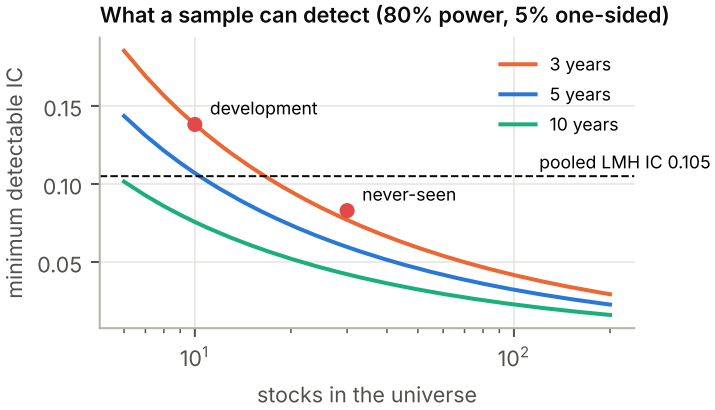


Figure 17. Minimum detectable mean IC (80% power, one-sided 5%) by universe size and sample length. Red dots: the two universes of this paper. Dashed: the pooled LMH IC.

What converts it into evidence is that the holdout and the never-seen period, which played no part in the search, returned the same information coefficient.

Power: With N stocks and M independent monthly cross-sections, the smallest detectable mean IC at 80% power is $(z_{0.95} + z_{0.80})/\sqrt{(N-1)M}$ (Figure 17). That is 0.138 for ten stocks over three years, 0.077 for thirty, and 0.032 for a hundred stocks over five years. A single ten-stock sample therefore has a coin-flip chance (0.56) of detecting an IC of 0.10, so a null result in one would be uninformative. The never-seen period raises power to 0.91, and the three samples pooled exceed 0.99. Monte Carlo on the calibrated generator gives the same message: with a true effect of the size in the data, the full pipeline detects it in 0.45 of ten-stock five-year worlds and 0.70 of thirty-stock worlds.

11 Path risk: generative stress testing

The Calmar ratio is the objective of a concentrated manager, and it is the statistic that small samples measure worst. A five-year history contains one or two drawdown episodes, so the maximum drawdown, and hence the Calmar ratio, is close to a single draw from its

distribution. Recovering that distribution requires many plausible histories that preserve what matters for drawdowns: fat tails, volatility clustering, and the nonlinear co-movement of the strategy with the market and with momentum, its closest factor relative.

Four generators: I train generators on the joint daily returns of FMF-N, the equal-weight market and the 12–1 momentum book, pooling the development period and the never-seen period after standardising each by its own mean and volatility (≈1,900 days).

- The *stationary block bootstrap* (Politis and Romano, 1994) re-orders history.
- *Filtered historical simulation* (Barone-Adesi, Giannopoulos, and Vosper, 1999) re-uses history's joint shock vectors inside GJR-GARCH-*t* volatility dynamics.
- *TimeGAN* (Yoon, Jarrett, and van der Schaar, 2019) learns the whole path distribution adversarially.
- The fourth generator, *Diffusion-FHS*, is a hybrid designed for small samples. GJR-GARCH-*t* with variance targeting carries the volatility dynamics, where five years of data suffice. A conditional denoising diffusion model (Ho, Jain, and Abbeel, 2020), with a residual-MLP denoiser, learns the joint law of the daily shock vector *given the volatility regime*. That is where flexibility pays and where 1,900 observations of a three-dimensional vector suffice. It can therefore produce shock combinations that history contains rarely, for example a strategy loss that coincides with a momentum crash in a high-volatility state, instead of replaying the few that occurred. Because Gaussian-noise diffusion under-represents heavy tails when data are scarce, the shocks are mapped to normal scores before training, the Gaussian quantile transform of tabular diffusion (Kotelnikov et al., 2023), and mapped back through the empirical quantile function with Student-*t* tails beyond the sample. The diffusion model thus learns the regime-dependent *dependence*, a conditional copula, while the marginal tails come from the data.

The denoiser is selected by early stopping on a chronologically held-out period. An unregularised network trained on this little data memorises its training set, and its draws collapse towards it. Sampling uses a strided DDIM-type chain (Song, Meng, and Ermon, 2021). Per-day moment matching (Boyle, Broadie, and Glasserman, 1997) removes the small first- and second-moment biases of the sampler, which would otherwise compound into a spurious five-year drift. The GARCH recursion uses variance targeting (Engle and Mezrich, 1996), with persistence capped at 0.99.

Fidelity: Table 18 reports a common battery, including a discriminative score (a gradient-boosting classifier trained to separate real from synthetic 64-day windows, with real windows split chronologically so that overlapping twins cannot leak) and a memorisation ratio. The block bootstrap matches the marginal moments by construction (strategy excess kurtosis 5.3 against 5.5), but it can only re-order the days that happened. Diffusion-FHS has the lowest discriminative score of all generators (0.014, so the classifier can barely tell its windows from real ones), a memorisation ratio of 0.96, so it does not copy, and a correlation with momentum of 0.62 against 0.66. Its tails are heavier than the sample's (strategy excess kurtosis 10.2 against 5.5; market 34 against 13): the GARCH recursion compounds rare large shocks into longer

Table 18. Generative models: fidelity and path risk (pooled standardised samples; 1,000 five-year paths each)

	FMF kurt.	MKT kurt.	corr(FMF, MOM)	Discrim.	Memoris.	Median Calmar	5% MaxDD (%)
Real data	5.5	13	0.66	–	–	–	–
Block bootstrap	5.3	13	0.67	0.067	0.97	0.91	−26
FHS (GJR-GARCH-t)	9.8	63	0.56	0.019	1.02	0.84	−29
Diffusion-FHS	10.2	34	0.62	0.014	0.96	0.70	−34
TimeGAN	−0.5	-1	0.99	0.500	1.11	1.26	−20

Discriminative score: |accuracy−0.5| of a classifier separating real from synthetic windows (0 = indistinguishable). Memorisation ratio: nearest-training-window distance of synthetic windows relative to held-out real windows (1 = no copying). Calmar and drawdown use the posterior-predictive Sharpe ratio as drift.

high-volatility spells than five years happened to contain. For a stress test this errs on the side of caution, and FHS errs much further (market excess kurtosis 63). TimeGAN fails: its draws are Gaussian-looking and nearly collinear across channels (correlation with momentum 0.99), and a classifier separates them perfectly (score 0.500). This failure mode of adversarial training on short, noisy series is reported rather than tuned away, and TimeGAN is excluded from inference.

Drift is an input, not an output: No generator can learn a mean from five years of data. Paths are generated with zero drift, and the strategy's drift is drawn from the posterior-predictive distribution of its Sharpe ratio in a new universe (Section 10.5). Parameter uncertainty and path risk therefore enter the Calmar distribution together. Under Diffusion-FHS the median five-year Calmar ratio is 0.70, the probability of a Calmar ratio above 0.5 is 59%, the median maximum drawdown is −15% and its 5% quantile is −34% (Figure 18). The probability of losing money over five years is 14%. These are the numbers to put in front of an allocator, rather than the single in-sample Calmar ratio.

Conditional stress: The binding risk is momentum, not the market. In simulated three-month windows where the equal-weight market draws down by 25% or more, FMF-N's mean return is positive under every generator (6.4%, 0.6% and 10.9% for the bootstrap, FHS and Diffusion-FHS). In the 2% of windows with the worst momentum returns, the 5% quantile of the three-month return falls to between −14.6% and −16.8% (bootstrap −14.1%), roughly twice the unconditional −8.3%. The three generators agree on this tail even though they build it differently: the bootstrap replays the 2020–21 momentum reversals, FHS rescales them through volatility dynamics, and Diffusion-FHS draws new momentum-correlated shocks in high-volatility states. Agreement across mechanisms is the evidence that the scenario is a property of the strategy, not of one model. For a PM, the practical consequence is a momentum hedge or a cap on the book's momentum exposure in high-volatility regimes. For a QIS structurer, this is a stress scenario that we disclose.

12 Discussion: implications for practitioners

What is robust: Across the development sample, the holdout and a later, broader universe, four facts recur:

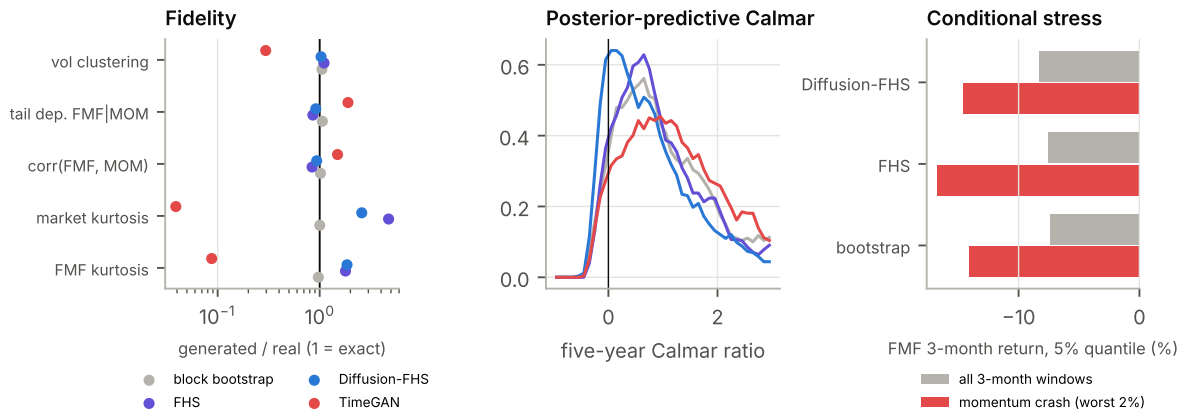


Figure 18. Left: stylised facts of real and generated data. Centre: distribution of the five-year Calmar ratio with posterior-predictive drift. Right: 5% quantile of FMF-N's three-month return, unconditionally and in simulated momentum crashes.

- a zero-beta book built on revision flow earns a Sharpe ratio near one after costs
- its alpha survives the market and the price factors
- leader minus herd predicts returns with an IC near 0.10
- the book improves almost any institutional strategy it is blended with, because its correlation with them is moderate and its crisis behaviour is defensive.

What is not: The specific construction cell is not identifiable (PBO 0.945, and 0.76 across the multiverse). The development Sharpe ratio alone does not survive deflation over the full research history, and on any single short window a family-wise test over all specifications is not significant. The circular time-shift placebo is passed only in the holdout. The causal decomposition of Section 10.1 shows that this reflects the placebo's low power rather than a hindsight tilt, but the placebo result stands as reported. In development, P&L is concentrated in a few names and days. Momentum crashes are the binding stress. These are properties of concentrated universes and of a revision-based signal, and they are the reasons the validation protocol is needed.

For portfolio managers: FMF is a zero-beta selection sleeve. Its returns are defensive in market crashes and exposed to momentum crashes. Three implementation choices follow.

- **Size it on path risk:** The posterior-predictive drawdown distribution (5% quantile −34% over five years) is a more realistic budget than the in-sample drawdown.
- **Make beta a decision:** Beta-neutral legs or a disclosed beta budget are one-line choices in the package.
- **Watch the signal's health:** The realised rolling IC is known in real time and is the early warning of decay.

The breakeven cost of 234 bp per unit traded and turnover of two to three times a year leave ample room for execution in liquid large caps.

For QIS structurers: The strategy is rules-based, uses end-of-day data only, and re-targets one forty-second of the book each day.

- **Index delivery:** It can be delivered as an excess-return index with a volatility target; the package computes index levels, fees and a rulebook from the exact parameters.
- **Blending:** As a sleeve, a 50/50 combination raises the Calmar ratio of every price-factor index studied, in sample and out of sample.
- **Disclosure:** The momentum-crash scenario of Section 11 is the stress scenario to disclose.

For researchers: Three lessons travel beyond this strategy.

- In small universes, report what a sample *could* detect before reporting what it did.
- Replace the single backtest with the family of reasonable backtests and a selection-free ensemble.
- Combine independent samples formally, through a random-effects meta-analysis and its predictive distribution, rather than counting p-values below 0.05.

Implementation:

- **Inputs and execution:** The strategy is fully rules-based. It uses end-of-day prices and target prices with a one-session knowledge lag, and re-targets one forty-second of the book each session, which spreads execution.
- **Exposure and turnover:** Gross exposure is about 0.9 with zero net exposure, and turnover is two to three times per year.
- **Capacity:** Capacity is governed by the liquidity of the names held. No volume data are used, but the breakeven cost of 234 bp per unit traded leaves a wide margin for market impact in large caps.
- **Beta and blending:** Market exposure is a separate, explicit decision. With a constant beta budget of 0.2, the Calmar ratio over 2017–2021 rises to 1.66, above RTB's 1.39, with smaller drawdowns, without any timing claim. For a QIS structurer the book is a natural low-correlation sleeve, and a 50/50 combination improves the drawdown profile of every price-factor index studied.

Limits: Both universes are small and mostly cover rising markets, and the out-of-sample window is short. The out-of-sample universe, built from the public Kaggle data described in Section 4, adds twenty of today's large caps to the ten development names: the period is unseen, but the list was chosen knowing which companies grew large. The causal timing decomposition and the twenty-new-names check guard against the most likely bias; a universe defined by an independent rule, such as index membership at the start of the window, would be stronger evidence. Factor exposures beyond the market and four price factors (size, value, sector) cannot be measured without additional data, the package's sector-neutral option addresses the last of these for users who have a sector map. The natural next steps are a broad universe with analyst coverage, where the detectable IC falls to about 0.032, and a live, frozen forward test of the published specification.

13 Conclusion

Concentrated portfolios cannot rely on breadth, and their reported Calmar ratios are easily inflated by market exposure. Separating the two leads to a different research agenda: find a selection signal that adds to price momentum, harvest it with a construction matched to its information horizon, treat beta as a disclosed budget, and validate with tests that respect the sample's limited power. Following this agenda, First-Mover Flow turns the timing of analyst revisions into a zero-beta book that improves on its regime-timed predecessor, on the institutional toolkit and on blends with it, and that replicates in a period it had never seen, on twenty names it had never seen. Attacked with the objections a sceptical committee would raise, the result holds where it should:

- its source is timing rather than hindsight
- it survives realistic implementation
- its sign does not depend on the specification
- pooled across independent samples, the probability that the mechanism works in the next universe is about 0.96.

It is fragile where a revision-flow strategy should be: in momentum crashes, and in the precision of any single short sample. The ledger of what failed along the way, and the code that lets anyone rerun every test on their own universe, are part of the evidence.

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A Data cleaning log (development universe)

B Signal and comparable definitions

All signals are oriented so that higher is more attractive and use data known at t. *Price*: 12-1 momentum $\ln P_{t-21} - \ln P_{t-252}$; 6-1 momentum $\ln P_{t-21} - \ln P_{t-126}$; reversal $-(\ln P_t - \ln P_{t-21})$;

Dataset	Check	Action	Rows
Prices	clean	Non-sessions removed (coverage $< 50\%$)	55
Ratings	R1	Unparseable / non-positive target	9
Ratings	R2	Missing analyst identifier	47
Ratings	R3	TP/P outside (0.2, 5.0)	2
Ratings	R4	Target change inconsistent with price change (excess outside (0.33, 3.0))	0

Over the never-seen period the same rules remove eight ratings published after the last session and two implausible targets.

low volatility $-\mathrm{sd}_{60}(r)$; betting-against-beta $-\hat{\beta}_{252}$ on the equal-weight market; residual momentum: the 231-session sum of market-residual returns divided by their standard deviation, lagged 21 sessions; 52-week high $P_t / \max_{252} P$. *Analyst*: consensus upside $\ln(\text{median } TP/P)$; target-price momentum $\Delta_{63} \ln(\text{median } TP)$; low dispersion $-\mathrm{sd}(\ln TP)$; classic breadth $B_{i,t}$. *Blends*: equal-weight ranks of (12-1 momentum, low volatility, reversal) and of (12-1 momentum, breadth). *Allocations*: equal-weight and inverse-60-day-volatility long-only; time-series momentum (long or short each stock by the sign of its 12-1 return, rebalanced monthly).

C Pre-registrations

Each audit round's hypotheses and the out-of-sample transfer rules were serialised to JSON and hashed (SHA-256, first 16 hexadecimal characters) before outcomes were computed. The out-of-sample file (`configs/oos_open_universe_preregistration.json`, hash `0b2b0792fd357d61`) is verified by the notebook before any result is printed. It fixes the frozen parameters, the primary and secondary leg rules, the warm-up, the evaluation window, the test list, and the rule that nothing is changed afterwards.

D Development universe identification

By the author's reconciliation of return signatures against public closing prices, the ten development series correspond to large US technology, consumer, industrial and financial names (stock_1 AAPL, stock_2 BA, stock_3 CHGG, stock_4 INTC, stock_5 JPM, stock_6 MSFT, stock_7 NKE, stock_8 NVDA, stock_9 PANW, stock_10 QCOM). The identification is used only for interpretation; no rule uses it.

E Generative models: specification

Diffusion-FHS: For each channel c (FMF-N, equal-weight market, 12–1 momentum book), standardised returns $x_{c,t}$ follow a GJR-GARCH(1,1) recursion $\sigma^2_{c,t+1} = \omega_c + (\alpha_c + \gamma_c \mathbb{1}_{e_{c,t}<0}) e^2_{c,t} + \beta_c \sigma^2_{c,t}$, $e_{c,t} = \sigma_{c,t} z_{c,t}$. The parameters (α, γ, β) are estimated by Student-t quasi-likelihood on the pooled samples, persistence is capped at 0.99, and ω is set by variance targeting. A conditional DDPM learns $p(u_t \mid c_t)$, where $u_t \in \mathbb{R}^3$ are the normal scores of the innovation vector z_t (rank-based Gaussian quantile transform) and c_t the vector of standardised log volatilities. Draws are mapped back to z_t through the empirical quantile functions, extended beyond the sample by standardised Student-t tails with the GARCH degrees of freedom. The denoiser is

a residual MLP: three blocks of width 128, SiLU activations and a sinusoidal time embedding. It uses a cosine schedule with $T = 100$ steps (Nichol and Dhariwal, 2021), the simplified ϵ-prediction loss, Adam with warm-up and cosine decay, an exponential moving average of the weights (0.999), and early stopping on the last 15% of each sample. Simulation alternates the GARCH recursion and a 50-step strided ancestral sampler (Song, Meng, and Ermon, 2021) conditioned on the current volatility state. Shock vectors are moment-matched across paths each day (zero mean, empirical standard deviation), and volatility states are kept within the observed range widened by half a standard deviation.

Baselines: The stationary bootstrap uses a mean block of 21 days. FHS uses the same GJR-GARCH dynamics with jointly resampled empirical residual vectors. TimeGAN follows Yoon, Jarrett, and van der Schaar (2019) with single-layer GRUs (hidden size 24), 24-day windows, and 1,500 embedding, 1,500 supervised and 2,000 joint iterations.

Fidelity battery: The battery reports moments, the autocorrelation of absolute returns (lags 1–10), cross-correlations and 5% tail dependence. It also reports a discriminative score from a gradient-boosting classifier on window features and raw values, with real windows split chronologically and a one-window gap, and a memorisation ratio based on Euclidean nearest-neighbour distances to training windows relative to held-out real windows.

F Software architecture

The package `fmflow` is organised by responsibility:

- `data` holds the contract, cleaning with an audit log, and the point-in-time snapshot
- `signals` holds the registry of built-in and user signals
- `risk` holds the walk-forward GJR-GARCH and an on-disk cache
- `portfolio` handles legs, allocators, beta neutrality, caps, volatility targeting and tranches
- `backtest` handles costs, borrow and execution delay
- `analytics` covers metrics, inference, snooping-robust tests, Bayesian tools, attribution and power
- `robustness` runs placebos, leave-one-out, the multiverse and truncation checks
- `generative`, `production` (book, trades, risk report, signal monitor) and `index` (QIS index levels and rulebook) complete the package.

Every tunable number lives in two frozen configuration classes that serialise to JSON and can be fingerprinted before evaluation. A command-line interface exposes the workflows. Ten notebooks cover data ingestion, signal research, construction and risk, robustness and inference, generative stress testing, QIS index construction, the PM's daily workflow and a complete pre-registered research cycle on a user's own universe.

G Reproducibility

The accompanying repository (`first-mover-flow`) contains a self-contained Python package, `fmflow`. It implements the full pipeline from two raw tables (daily prices; analyst target prices with publication dates) to daily weights and net returns, organised in modules for the data contract and audited cleaning, the point-in-time snapshot, signals, GJR-GARCH risk, portfolio construction and the backtest. The validation battery of this paper is included: phase-averaged IC tests, PSR/DSR, CSCV, placebos through the identical construction, leave-one-out rebuilds, the 13 institutional comparables and three allocations, and a pre-registration utility that fingerprints a specification before evaluation. A structural synthetic data generator is calibrated to both universes and is used to check that the pipeline finds no alpha in a null world. More than 100 unit, causality (truncation) and parity tests run in continuous integration on Linux and Windows. With the notebook's GARCH forecasts, the package reproduces the research notebook's weights exactly on the development universe (maximum absolute difference 0) and to within 5×10^{-13} on the open universe. Re-estimating GARCH from scratch is exact on one machine but not across machines: on a few hundred observations the GJR-GARCH-t likelihood has near-equal local maxima, and a gradient optimiser's stopping point depends on the processor's floating-point path (a different BLAS kernel moves the out-of-sample Sharpe ratio from 1.05 to 1.02). The package therefore also offers a deterministic grid-likelihood estimator with variance targeting, which is bit-identical across machines and gives Sharpe ratios of 1.09 out of sample and 1.12 over 2017–2021. The open data, the notebook that rebuilds them from their public sources, and the registration file are distributed; the proprietary development data are not.